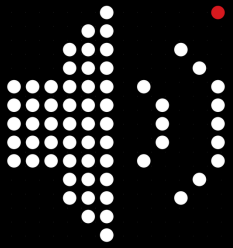
CURRENTLY IN EXPERIMENTAL BETA AND IN TESTING MODE



CMF STEREO

THE APP FOR CMF TRUE STEREO: REAL STEREO ON THE CMF PHONE 1, NO TERMINAL NEEDED

ON AND OFF • MONO AND STEREO • EARPIECE AND SPEAKER LEVELS
SPEAKER BOOST • DEEP BASS GUARD • EARPIECE LIMITER AND LEVELLER
A STEREO SPEAKERS PAGE INSIDE NOTHING SETTINGS, FOUND BY SETTINGS SEARCH
A QUICK SETTINGS TILE • NOTHING-STYLE EDITION AND A PRO EDITION

©DR.ESSENCE • NDOT FONTS ARE OWNED BY ©NOTHING

INSTRUCTION MANUAL

V1.0.97-BETA • MANUAL IN EARLY BETA

USER MANUAL FOR CMF STEREO AND CMF STEREO PRO, THE APPS FOR
THE CMF TRUE STEREO MODULE ON THE CMF PHONE 1

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WELCOME

Thank you for using CMF Stereo. This chapter tells you what the app is, how it relates to the module it controls, which of the two editions suits you, what you need before you start, and the few safety rules that matter most.

1.1 About this manual

This manual covers version v1.0.97-beta of both editions of the app: CMF Stereo (the everyday edition) and CMF Stereo Pro (the edition for tinkerers). It also covers the small companion app CMF L/R Test, which ships with every release.

Each topic is first explained in plain words. Where precise detail helps, it follows in a marked box:

> TIP
>
> **Note** boxes give tips and useful background.

> WARNING
>
> **Warning** boxes are about the safety of your speaker, your earpiece or your phone. Please read every one.

> TECHNICAL
>
> **Technical** boxes give exact values, file names and commands, for readers who want to know what happens underneath.

On-screen labels are written exactly as the app shows them. CMF Stereo writes its labels in capitals (for example EARPIECE SPEAKER); CMF Stereo Pro uses normal sentence case (for example Earpiece output).

1.2 What CMF Stereo is

The CMF Phone 1 has two sound outlets on its front and bottom: the **earpiece** at the top (the one you hold to your ear in a call) and the **bottom speaker**. Out of the box, media plays from the bottom speaker only.

The **CMF True Stereo** module makes the earpiece play media too. In **Mono** both play the same full sound; in **Stereo** they become a real left and right pair, and left and right follow the screen however you turn the phone.

CMF Stereo is the app for that module. It lets you control the module with taps instead of terminal commands: switch the earpiece on and off, choose Mono or Stereo, set the earpiece and speaker levels, and set the module's stereo protection.

> TIP
>
> The module works perfectly well on its own. The app is the optional, friendly front end: everything it does is a call to the module's own command, `stereoctl`, or an edit to the module's settings files. The module stays the source of truth, and the terminal keeps working exactly as before.

1.3 The module and the app

Two separate pieces work together:

The module: CMF True Stereo	The app: CMF Stereo / CMF Stereo Pro

What it is	A Magisk / KernelSU / APatch module (a zip file)	An ordinary Android app (an APK file)
How you install it	Flash it in your root manager, then reboot	Install it like any app
What it does	All the real work: routing the sound to the earpiece, Mono and Stereo, protection	Shows the module's state and changes its settings
Needed?	Yes	Optional

Because the app only talks to the module, it can do nothing without it. Install the module first.

> TECHNICAL
>
> The app reads and writes through a root shell: the module's command line at `/data/adb/modules/cmf_stereo/scripts/stereoctl`, and its settings in `/data/adb/cmf-stereo/stereo.conf` and `/data/adb/cmf-stereo/actions.conf`. The module's daemon reads its configuration every cycle, so changes take effect without restarting anything.

1.4 Two editions

Both editions are built from the same code, drive the same module and can be installed side by side.

	CMF Stereo	CMF Stereo Pro

Who it is for	Everyone, for everyday listening	Tinkerers, for tuning and diagnosing
Look	Nothing's design language: black, NDot type in glowing green	Material dark
Controls	On/off, Mono/Stereo, earpiece and speaker level on a plain 1 to 10 scale, speaker boost, deep bass guard, earpiece leveller and limiter	The same, in raw values (dB), plus status details and Pro-only tools
Developer tools	None	Mixer controls, diff, probe, sweep, logs, config files, and a built-in guide
Quick Settings tile	Stereo	Stereo Pro
Release file	CMF-true-stereo-clean-….apk	CMF-true-stereo-pro-….apk

Only CMF Stereo Pro has: the detailed status card, the Mode setting (While playing / Always on), Solo and Un-solo, Doctor, the earpiece gain slider in dB, the stereo attenuation field in dB, the daemon settings, the Danger zone, the developer tools and the Guide.

1.5 Which edition is for you

- Choose CMF Stereo if you want to switch stereo on, set two levels and listen. Every control is a simple tap, and the speaker can never be set below its safety

margin.

- Choose **CMF Stereo Pro** if you want to see exactly what the module is doing, set values in decibels, or diagnose a problem.
- Install both if you like: use CMF Stereo day to day and keep Pro for the occasional look underneath. They do not get in each other's way, because both read the same state from the module.

> TIP
> If you also want the Stereo speakers page inside Nothing's Settings (chapter 6), enable only one edition in LSPosed.

1.6 What you need

- A **CMF Phone 1 (A015)**. On any other phone the module installs and does nothing.
- **Root** through Magisk, KernelSU or APatch, and superuser permission granted to the app.
- The **CMF True Stereo module**, flashed and the phone rebooted. Mono/Stereo needs module v0.9.0 or newer; the protection settings need v1.0.4 or newer; the earpiece leveller needs v1.0.5 or newer; instant updates of the Nothing Settings page for changes made outside the app need v1.0.6. This release of the app is made for module v1.0.6 or newer; older modules show fewer settings.
- **Android 8.0** or newer.
- The app installed on **internal storage** (it refuses to install anywhere else).

- Optional, for Stereo inside Nothing Settings and Settings search: **LSPosed**.

> TECHNICAL
>
> Tested phone: CMF Phone 1 (A015), Nothing OS 4.1 Tetris-B4.1-260615-1652-EEA, Android 16, with module v1.0.6. Other regions (Global, India) are untested: the module checks the audio files by hash and only patches the exact ones it was built for, so on a build whose files differ it refuses Stereo and stays in Mono.

1.7 Safety first

Please read this before you use Stereo.

> WARNING
>
> **The speaker in Stereo.** In Mono, Awinic's speaker protection (the phone's own) keeps the speaker within its limits from a live model of the speaker. Stereo has to switch that processing off for media. In its place, the module holds the speaker 6 dB below the amp's full output at its loudest, holds back its deepest bass when it gets loud, and relies on the amp's own hardware cut-outs. These are fixed limits: there is **no temperature model.** For long, loud listening, use **Mono.** In Stereo, lower the speaker level and back off at the first distortion.

WARNING

The earpiece. The earpiece is a receiver, built for calls. It has no protection circuit, no excursion limit and no thermal cut-off. Distortion, buzz or rattle means turn it down at once: that damage is permanent, and it also costs you the earpiece for phone calls. If you play it loud for hours, check whether the top of the phone gets warm.

- Calls always keep Awinic's protection, in Mono and in Stereo.
- CMF Stereo never sets the speaker below its 6 dB margin. CMF Stereo Pro can, but only after a clear warning.
- If the phone fails to boot twice in a row with Stereo on, the module removes Stereo by itself and falls back to Mono.

1.8 Mono and Stereo in one page

	Mono	Stereo
What you hear	Both speakers play the same full sound	The earpiece and the bottom speaker play left and right
Speaker protection	Awinic's full protection, as stock	The module's own protection and a fixed attenuation
Best for	Loud listening	Games, films and music with real sides
Default	Yes	No

Left and right follow the screen. Hold the phone sideways with the earpiece on your left and the earpiece plays the left channel; turn it the other way and it plays the right. Stereo is never reversed (module v0.9.5 or newer).

Switching is instant. With module v0.9.4 or newer, the module restarts the phone's audio service: the sound stops for about a second, and you press play again. Only when that cannot be done safely does it ask you to restart the phone.

1.9 Words used in this manual

Word	Meaning
Earpiece	The small speaker at the top of the phone, used for calls. Also called the receiver.
Speaker	The bottom speaker, driven by an Awinic smart amplifier.
Armed / disarmed	Whether the module plays media through the earpiece (EARPIECE SPEAKER on or off).
Daemon	The module's small background program that keeps the earpiece routing applied.
Audio HAL	The phone's audio service (MediaTek's). The module restarts it to switch Mono and Stereo.
Awinic protection	The speaker protection processing that runs on the phone's audio DSP in Mono.
Protection library	The module's own protection (bass guard, limiters, boost, leveller), which runs inside the audio HAL in Stereo.

dB, dBFS	Decibels. dBFS is relative to digital full scale (0 dBFS is the loudest a signal can be).
Staged / pending	A Mono/Stereo choice that is saved but waits for a restart before it runs.
LSPosed	A framework that lets an app change other apps (here: Nothing's Settings). Optional.

02

INSTALLING, UPDATING AND REMOVING

Installing takes two steps: the module first, then the app. This chapter also explains where to get the files, how updates work, and how to remove the app or the module again.

2.1 Before you start

- Make sure your phone is rooted with Magisk, KernelSU or APatch and that the root manager opens normally.
- Download the module zip from the CMF True Stereo releases page (repository `Dr-Essence/CMF-true-stereo`).
- Download the app from the app's releases page (repository `Dr-Essence/CMF-true-stereo-manager`), or use Komi Store (see below).

> TIP
> Every module release lists the SHA-256 checksum of its files, in the release description and in `SHA256SUMS.txt`, so you can check your download.

2.2 Step 1: flash the module

1. Open Magisk, KernelSU or APatch and go to its Modules page.
2. Install the module zip from storage.
3. Reboot the phone.

After the reboot the module runs in Mono, its default. You can already switch to Stereo without the app: tap **Action** on the module's card in your root manager, or run `su -c stereoctl channels stereo` in a terminal.

> TIP
>
> The module is a Magisk zip, not an app, so it is not offered in app stores such as Komi Store. Your root manager offers new module releases as updates.

2.3 Step 2: install the app

1. Open the APK of the edition you want: `CMF-true-stereo-clean-….apk` for CMF Stereo, `CMF-true-stereo-pro-….apk` for CMF Stereo Pro.
2. Allow installing from that source if Android asks.
3. Open the app. Your root manager asks whether to grant it superuser access: allow it.

The app then reads the module's state and shows its controls.

SCREENSHOT: Magisk superuser request dialog asking to grant root to CMF Stereo, on first launch

> TECHNICAL
>
> Package names: CMF Stereo is `com.tiimoo.cmfstereo.clean`, CMF Stereo Pro is `com.tiimoo.cmfstereo`. Each release APK is about 1 MB.

2.4 Granting root

The app needs root because the module's files and command live in the protected `/data/adb` folder. Without root it can only show a message (NEEDS ROOT in CMF Stereo, No root access in Pro).

If you refused the request by mistake, open your root manager's Superuser list, allow CMF Stereo (or CMF Stereo Pro), and reopen the app.

> TIP
> Your root manager may show a short message (a toast) each time the app uses root. The app keeps these to a minimum: it reads everything it needs in a single root call, and the Nothing Settings page and Sound row read the last known state without root at all.

2.5 Both editions side by side

You can install CMF Stereo and CMF Stereo Pro at the same time: they have different package names. Both show the same state, because they both read it from the module. A change made in one appears in the other the next time it reads the state (in Pro, tap Refresh).

Each edition has its own Quick Settings tile (Stereo and Stereo Pro), so two installed tiles are easy to tell apart.

> WARNING
>
> For the Nothing Settings page, enable only one edition in LSPosed. With both enabled, Settings shows two Stereo speakers rows.

If both are installed, update both when a new release comes out.

2.6 Internal storage only

Both editions install on internal storage only. This matters for the Nothing Settings page: LSPosed starts before adopted storage (an SD card formatted as internal storage) is mounted, so it can never load an app that sits there.

If an override still put the app on the SD card, the app shows a panel that moves it back: **MOVE TO INTERNAL STORAGE** in CMF Stereo, **On adopted storage** in Pro. See chapters 3 and 4.

> TECHNICAL
>
> The manifest sets `android:installLocation="internalOnly"`. The developer option Force allow apps on external overrides it. The app detects adopted storage by its install path starting with `/mnt/expand/`; LSPosed then logs "Failed to find package info" at boot. The move runs `pm move-package <package> internal` as root, and the app restarts.

2.7 Staying updated with Komi Store

Komi Store is a free, open-source app store for apps released on GitHub. Search for CMF-true-stereo-manager there, install the edition you want, and Komi Store keeps it updated from the project's releases.

This works because every release is signed with the same key, so each update installs straight over the app you have.

2.8 Updating the app

Install the new APK over the old one, or let Komi Store do it. You do not need to uninstall first, and your settings stay as they were (they are stored by the module, not by the app).

- Releases from v1.0.35 on are signed with the project's permanent key and install over each other.
- From v1.0.95-beta on, versions are marked **beta**. The build number keeps rising, so v1.0.95-beta installed straight over v1.0.81, and app stores see each one as an update.
- If both editions are installed, update both.

> WARNING
>
> Builds older than v1.0.35 were signed with temporary keys. If you still have one, uninstall it once before installing a current release. A build you compile yourself, or one from a fork, is debug-signed and cannot update an installed release either.

2.9 Updating the module

Your root manager (Magisk, KernelSU or APatch) offers new module releases as updates; install the update and reboot. The module's settings in `/data/adb/cmf-stereo/` survive module updates.

Some app features depend on the module version. When an app release needs a newer module, its release notes say so (for example, v1.0.71 asked for module v1.0.6).

2.10 Which files are in a release

File	What it is
`CMF-true-stereo-clean-v<version>-<commit>.apk`	CMF Stereo, the everyday edition
`CMF-true-stereo-pro-v<version>-<commit>.apk`	CMF Stereo Pro
`CMF-LR-test-<commit>.apk`	CMF L/R Test, the left/right test app (chapter 7)

The newest tagged release is the stable one; earlier ones are kept as pre-releases for reference. There is also a rolling pre-release called Latest build, which moves with

every change to the code. For normal use, take the tagged release.

2.11 Uninstalling the app

Uninstall CMF Stereo or CMF Stereo Pro like any other app. The module keeps working with the settings you last chose, because those settings belong to the module, not to the app. Its Quick Settings tile goes with it, and so does its Stereo speakers row in Nothing Settings if you used LSPosed.

> TIP
> If you used the LSPosed part, you can also untick the app in LSPosed before uninstalling.

2.12 Uninstalling the module

Remove the module in your root manager and reboot. Its uninstaller stops the daemon and restores every audio control it touched.

The folder `/data/adb/cmf-stereo/` (the module's settings and routing) is left in place, so a reinstall does not lose your settings. Delete it by hand for a clean slate.

Without the module, the app shows MODULE NOT INSTALLED (CMF Stereo) or Module not installed (Pro).

2.13 If the phone does not boot

The module protects the boot by itself: if two boots in a row fail while the Stereo overlay is active, it removes Stereo and the phone boots in Mono. A bad boot costs one extra reboot instead of a trip to recovery.

If you ever need to switch the module off by hand:

```
adb wait-for-device shell
su
touch /data/adb/modules/cmf_stereo/disable
reboot
```

You can also hold volume down for safe mode. Both Magisk and KernelSU honour the `disable` file.

03

CMF STEREO, THE EVERYDAY APP

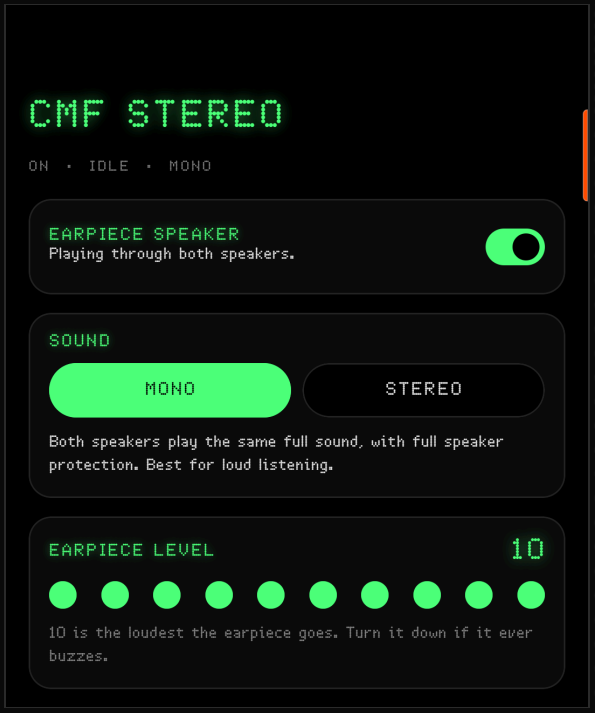
CMF Stereo is one scrolling screen of panels in Nothing's style: black, with NDot type in glowing green. This chapter goes through it from top to bottom, including every message it can show.

3.1 The screen at a glance

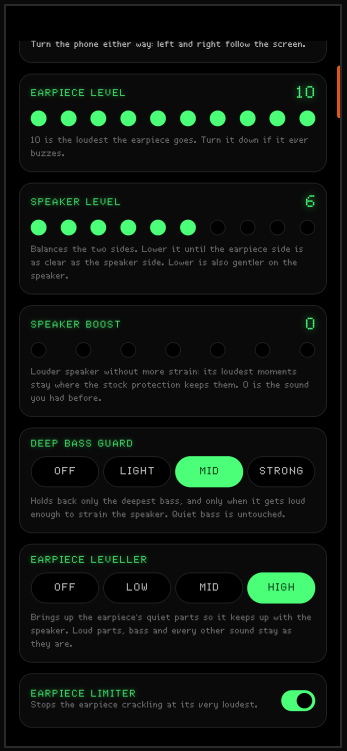
From top to bottom, CMF Stereo shows:

1. The title, CMF STEREO, and a status line under it.
2. Any notice that needs your attention: MOVE TO INTERNAL STORAGE, NOTHING OS UPDATED.
3. EARPIECE SPEAKER: on or off.
4. SOUND: MONO or STEREO.
5. EARPIECE LEVEL: 1 to 10.
6. In Stereo only: SPEAKER LEVEL, SPEAKER BOOST, DEEP BASS GUARD, EARPIECE LEVELLER and EARPIECE LIMITER.
7. A short confirmation of your last change, in dim green, at the bottom.

Scroll down to see all of it. Panels that do not apply right now are not shown at all.



The main screen in Mono: only the earpiece switch, Sound and the earpiece level.



Stereo, scrolled down: speaker level, boost, deep bass guard, leveller and limiter.

3.2 The title and the status line

Under the glowing CMF STEREO title, a grey line in capitals tells you the state at a glance. While the app reads the module it says READING…. Then it shows three words separated by dots, for example ON • PLAYING • STEREO:

Part	Values	Meaning
1	ON / OFF	Whether the earpiece is switched on (EARPIECE SPEAKER)
2	PLAYING / IDLE	Whether the module sees audio playing right now
3	MONO / STEREO	The channels the phone is running now

Instead of those three words, the line can say NEEDS ROOT or MODULE NOT INSTALLED (next section).

> TIP
>
> CMF Stereo reads the state when it opens and after every change you make. To see a change made somewhere else (Pro, the Quick Settings tile, a terminal), close and reopen the app.

3.3 NEEDS ROOT and MODULE NOT INSTALLED

If the app cannot use root, or cannot find the module, it shows a single panel instead of the controls:

Status line	Panel text	What to do
NEEDS ROOT	Grant this app superuser access, then reopen it.	Allow CMF Stereo in your root manager's Superuser list, then close and reopen the app.
MODULE NOT INSTALLED	Flash the CMF Stereo module in Magisk and reboot.	Flash the CMF True Stereo module zip in Magisk, KernelSU or APatch, and reboot.

SCREENSHOT: CMF Stereo showing NEEDS ROOT status line and the panel Grant this app superuser access, then reopen it.

SCREENSHOT: CMF Stereo showing MODULE NOT INSTALLED and the panel Flash the CMF Stereo module in Magisk and reboot.

> TECHNICAL
>
> NEEDS ROOT appears when the root shell does not start or does not run as user 0. MODULE NOT INSTALLED appears when /data/adb/modules/cmf_stereo/scripts/stereoctl does not exist.

3.4 MOVE TO INTERNAL STORAGE

This panel appears only if CMF Stereo has ended up on adopted storage (an SD card formatted as internal storage). It says, in red: CMF Stereo is on the SD card. Its Settings entry (through LSPosed) only works from internal storage.

Tap **MOVE**. The app moves itself to internal storage and restarts; the confirmation line says **MOVED**. If you use the Nothing Settings page, reboot afterwards so LSPosed can load it.

SCREENSHOT: CMF Stereo with the MOVE TO INTERNAL STORAGE panel and its MOVE pill (only shown when the app is on adopted storage)

> TIP
>
> Everything else in the app works from the SD card too; only the LSPosed part needs internal storage. This panel has not yet been seen on a phone, because the app normally refuses to install on the SD card.

3.5 NOTHING OS UPDATED

After a Nothing OS update, the phone's audio files may change. The module only patches files it knows, so it then keeps Stereo off. CMF Stereo tells you with a panel: *This Nothing OS version is new to the module, so Stereo is off and both speakers play the stock sound. Stereo comes back with a module update that supports it.*

There is nothing to set. Keep using Mono, and update the module when a release supports your new Nothing OS build.

SCREENSHOT: CMF Stereo with the NOTHING OS UPDATED panel in red text (only after a Nothing OS update the module does not know)

TECHNICAL

Shown when `stereoctl status` reports `build: updated` (module v0.9.8 or newer). Before anything is mounted at boot the module checks the vendor audio files against the analysed build; if they differ, the Stereo overlay is removed and the phone boots on stock Mono. This panel has not yet been seen on a phone.

3.6 EARPIECE SPEAKER

The main switch. On, media plays through the earpiece as well as the bottom speaker; the panel says Playing through both speakers. Off, the module leaves the earpiece alone; the panel says Bottom speaker only.

Tap the switch to change it. The confirmation line says ON or OFF. The choice survives a reboot: a switched-off module stays off until you switch it on again.

WARNING

In Stereo, switching the earpiece off stops the module's daemon, and the speaker amp goes back to its stock left slot on the next stream, so the speaker would play the left channel alone. If you want the earpiece off for a while, choose MONO first.

TECHNICAL

On runs `stereoctl on` (arm), off runs `stereoctl off` (disarm), which stops the daemon and restores every control it touched.

3.7 Not running right now

If the earpiece is switched on but the module's background program (the daemon) is not running, the EARPIECE SPEAKER panel adds a red line, Not running right now., and a RESTART pill.

Tap RESTART to start the daemon again; the confirmation line says RESTARTED. If the line comes back, see chapter 8.

SCREENSHOT: CMF Stereo EARPIECE SPEAKER panel with the switch on, the red line Not running right now. and the RESTART pill

> TECHNICAL
>
> RESTART runs `stereoctl restart`, which restarts the watcher daemon.

3.8 SOUND: MONO or STEREO

The SOUND panel has two buttons, MONO and STEREO. The filled one is your choice. Under them, a line explains it:

- MONO: Both speakers play the same full sound, with full speaker protection. Best for loud listening.
- STEREO: Left and right from the earpiece and the bottom speaker. Turn the phone either way: left and right follow the screen.

Tap MONO to switch to Mono at once. Tap STEREO and the app first shows what Stereo means (next section).

When you switch, the sound stops for a moment and music players pause: press play again.

> TIP
>
> The SOUND panel only appears with module v0.9.0 or newer. On older modules there is no Mono/Stereo switch.

> TECHNICAL
>
> The buttons run `stereoctl channels mono` or `stereoctl channels stereo`. With module v0.9.4 or newer this applies at once by restarting the audio HAL on the wanted files. The selected button shows the chosen (staged) mode, which differs from the running one only while a switch waits for a restart.

3.9 The STEREO dialog

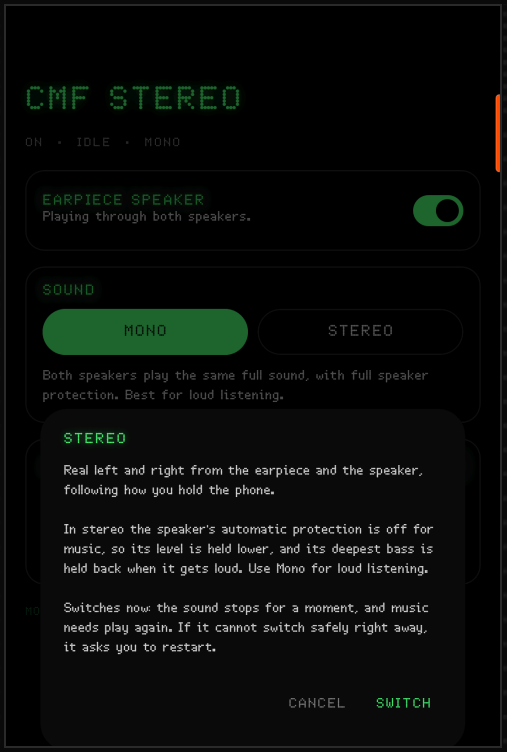
Before switching to Stereo, CMF Stereo shows a dialog titled STEREO:

Real left and right from the earpiece and the speaker, following how you hold the phone.

In stereo the speaker's automatic protection is off for music, so its level is held lower, and its deepest bass is held back when it gets loud. Use Mono for loud listening.

Switches now: the sound stops for a moment, and music needs play again. If it cannot switch safely right away, it asks you to restart.

Tap SWITCH to go to Stereo (the confirmation line says STEREO), or CANCEL to stay in Mono.



Tapping STEREO asks first: SWITCH or CANCEL.

3.10 Could not switch right away: RESTART PHONE

Sometimes the module cannot switch safely while the phone runs. It then saves your choice for the next start, and the SOUND panel says, in green: Could not switch right away. Restart the phone to switch to stereo. (or mono), with a RESTART PHONE pill.

Tap RESTART PHONE. A dialog titled RESTART says The phone restarts now to switch to stereo. (or mono). Tap RESTART to reboot now (the confirmation line says RESTARTING), or CANCEL to restart later yourself.

SCREENSHOT: CMF Stereo SOUND panel with the green line Could not switch right away. Restart the phone to switch to stereo. and the RESTART PHONE pill

SCREENSHOT: CMF Stereo RESTART dialog: The phone restarts now to switch to stereo., with RESTART and CANCEL

> TECHNICAL
>
> Reasons the module refuses a live switch: no verified files, a call in progress, or the audio HAL does not come back. The reboot runs `svc power reboot` (or `reboot`).

3.11 EARPIECE LEVEL

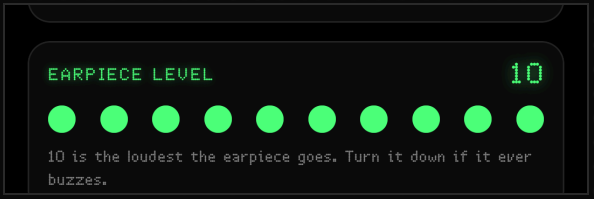
How loud the earpiece plays, from 1 to 10. The current level is shown as a large glowing number, and as a row of ten dots lit up to that level.

Tap a dot to set that level; the confirmation line says, for example, **EARPIECE 8.** The hint reads: 10 is the loudest the earpiece goes. Turn it down if it ever buzzes.

Level 10 is the earpiece amplifier's real maximum, the same value the phone itself uses in calls. Each level down is 2 dB quieter. The exact numbers are in chapter 5.

> **WARNING**
>
> If the earpiece ever buzzes, rattles or distorts, lower this level straight away.

> **TIP**
>
> This panel only appears when the app can read the earpiece gain from the module's settings. If it is missing, see chapter 8.



Earpiece level at 10, the loudest the earpiece goes.

3.12 SPEAKER LEVEL (Stereo only)

How loud the bottom speaker plays in Stereo, from 1 to 10. It appears only while Stereo is running.

The earpiece is smaller than the speaker and will not match it. Use this level to balance the two: lower it until the earpiece side sounds as clear as the speaker side. The hint reads: Balances the two sides. Lower it until the earpiece side is as clear as the speaker side. Lower is also gentler on the speaker.

Tap a dot to set it; the confirmation line says, for example, SPEAKER 7.

> TIP
> Level 10 is the safety margin itself: the speaker held 6 dB below the amp's full output. CMF Stereo can never go louder than that. Each level down is 2 dB more, down to 24 dB at level 1. In Mono this setting does not apply.

3.13 SPEAKER BOOST (Stereo only)

A louder speaker without more strain, from 0 to 6. The hint reads: Louder speaker without more strain: its loudest moments stay where the stock protection keeps them. 0 is the sound you had before.

The row has seven dots, for 0 to 6. Tap one to set it; the confirmation line says, for example, BOOST 3.

If the module's protection is not running yet, the hint starts with Starts with the next sound on the speaker. The boost only takes effect while the protection runs.

> TIP
>
> Shown with module v1.0.4 or newer, while Stereo is running. How it works: chapter 5.

3.14 DEEP BASS GUARD (Stereo only)

Four pills: **OFF**, **LIGHT**, **MID** and **STRONG**. The filled one is active. The text reads: Holds back only the deepest bass, and only when it gets loud enough to strain the speaker. Quiet bass is untouched.

Tap a pill to choose it; the confirmation line says, for example, **BASS MID**. The default is **MID**.

> WARNING
>
> The bass guard is part of what keeps the speaker safe in Stereo. Leave it at MID or STRONG unless you have a reason; OFF lets the deepest bass through at any level.

> TECHNICAL
>
> MID is the module's `medium` setting; the app runs `stereoctl protect bass off|light|medium|strong`.

3.15 EARPIECE LEVELLER (Stereo only)

Four pills: **OFF**, **LOW**, **MID** and **HIGH**. The text reads: Brings up the earpiece's quiet parts so it keeps up with the speaker. Loud parts, bass and every other sound stay as they are.

Tap a pill to choose it; the confirmation line says, for example, **LEVELLER LOW**. The default is **OFF**.

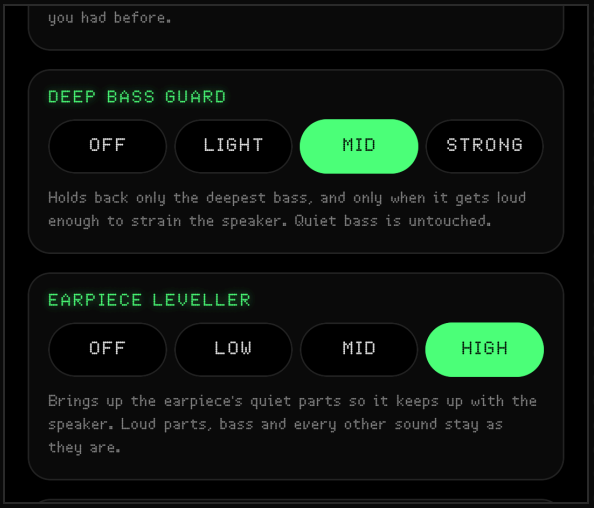
> TECHNICAL
>
> Shown with module v1.0.5 or newer. LOW, MID and HIGH are the module's light, medium and strong; the app runs stereoctl protect leveller ….

3.16 EARPIECE LIMITER (Stereo only)

A switch, on by default. The text reads: Stops the earpiece crackling at its very loudest.

Tap it to change; the confirmation line says **LIMITER ON** or **LIMITER OFF**.

> WARNING
>
> Keep the limiter on. Measurements showed the earpiece clips in the top 1.5 dB of its range; the limiter holds it below that.



Deep bass guard at MID, earpiece leveller at HIGH, the limiter on.

3.17 The confirmation line

After each change, a short word or two appears in dim green at the bottom of the screen: ON, OFF, RESTARTED, MONO, STEREO, RESTARTING, MOVED, EARPIECE 1 to 10, SPEAKER 1 to 10, BOOST 0 to 6, BASS OFF/LIGHT/MID/STRONG, LEVELLER OFF/LOW/MID/HIGH, LIMITER ON/OFF.

It tells you which change was sent. The panels above it then show the state the app read back from the module afterwards, which is the one that counts.

> TIP
>
> While a change is being made, every control is briefly disabled, so you cannot send two at once.

3.18 How the dots and pills work

- **Dots (levels):** lit dots show the level. Tap any unlit dot, or a lower lit one, to set that level. Tapping the dot you are already on does nothing. The value is written when you tap, not while you slide a finger.
- **Pills (choices):** the filled pill is the active choice. Tap another to choose it.
- **Switches:** green means on.

On SPEAKER BOOST, the first dot stands for 0 (off). At 0 no dot is lit; at 1 or more, the first dot lights along with the others.

3.19 When each panel appears

Panel	Shown when
MOVE TO INTERNAL STORAGE	The app is on adopted storage
NOTHING OS UPDATED	The module reports an unknown Nothing OS build (module v0.9.8+)
EARPIECE SPEAKER	Always (with root and the module)

SOUND	Module v0.9.0+
EARPIECE LEVEL	The earpiece gain can be read and is in range
SPEAKER LEVEL	Stereo is running (module v0.9.0+)
SPEAKER BOOST, DEEP BASS GUARD, EARPIECE LIMITER	Stereo is running and the module has protection (v1.0.4+)
EARPIECE LEVELLER	As above, and the module has the leveller (v1.0.5+)

> TIP
> "Running" matters: while a switch to Stereo waits for a restart, the Stereo-only panels stay hidden until the phone really runs in Stereo.

3.20 The look: NDot and the icon

CMF Stereo uses Nothing's own NDot dot-matrix font, taken from your phone at runtime: no Nothing font ships inside the app. It prefers the JP Extended cut (`Ndot77JPExtended.ttf` on the CMF Phone 1), then NDot 57, then any NDot it finds. On a phone without NDot it falls back to a monospace font.

The app icon is a speaker with two sound waves drawn as a dot matrix, with Nothing's red dot, on black. It has a themed (one-colour) layer for Nothing Launcher's Nothing icon style and Android 13+ themed icons. With Nothing2Something (formerly Nothing Launcher+) active on Nothing Launcher, the icon's sound waves pulse out from the speaker while it is on screen.

TECHNICAL

The font is looked for in `/system/fonts`, `/product/fonts` and `/system_ext/fonts`, then in Nothing's widgets app (`com.nothing.hearthstone`). It is copied into the app's own memory before use.

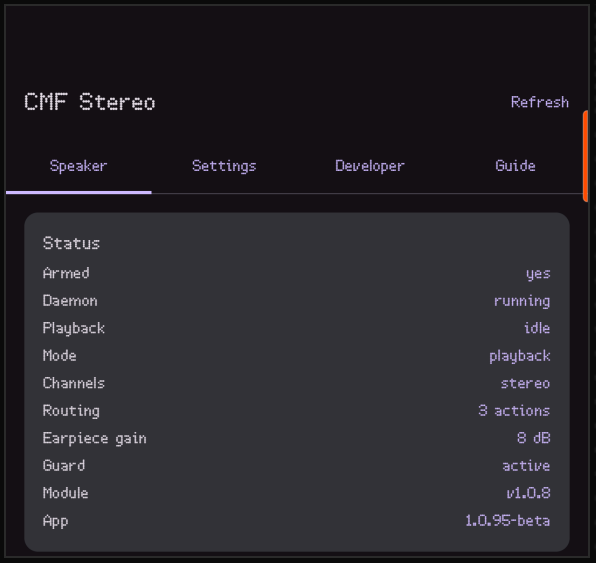
04

CMF STEREO PRO

CMF Stereo Pro shows everything the module does, in raw values, and adds the tools for tuning and diagnosing. It has four tabs: Speaker, Settings, Developer and Guide. This chapter goes through them card by card.

4.1 The screen at a glance

- A top bar with the title **CMF Stereo** and a **Refresh** button.
- A progress bar while the app works, with a line saying what is running.
- Four tabs: **Speaker** (everyday controls and status), **Settings** (the module's daemon), **Developer** (tools) and **Guide** (a searchable reference).
- An **Output** card at the bottom of every tab, showing what the last command printed.



Pro, Speaker tab: the Status card.

4.2 Top bar, Refresh and the progress line

Tap **Refresh** to read the module's state again, for example after a change made elsewhere. It is disabled while the app is busy.

While a command runs, a progress bar appears and every control is greyed out. For slow commands the app says what it is doing:

- Running: probe - dumping every mixer control, /sys, dmesg and the param XMLs. This takes a minute or two.
- Running: report - folding the newest probe into one file.
- Running: diff - snapshotting, waiting, then comparing. Start or stop playback now.

Other commands show Running: followed by their name, for example Running: channels.

4.3 No root access and Module not installed

Without root or without the module, Pro shows a red card instead of the tabs:

Card	Text
No root access	Grant superuser permission, then Refresh.
Module not installed	Flash CMF-true-stereo and reboot.

Fix the cause, then tap **Refresh**.

SCREENSHOT: CMF Stereo Pro red card No root access: Grant superuser permission, then Refresh.

4.4 The Output card

After every action, the **Output** card shows what the module printed, in a small monospace font. If a command printed nothing, it shows the action's name followed by : done. Tap **Clear** to hide it.

This is the place to look when a change seems not to work: the module's own message is there.

> TIP
>
> Values that the app writes itself are read back and checked. If a write did not stick, the Output card says so, for example FAILED to write gain: actions.conf still reads … or FAILED to write STEREO_ATTEN_DB: stereo.conf reads ….

4.5 Speaker tab: On adopted storage

If Pro has ended up on adopted storage, a red card titled On adopted storage appears at the top: Installed under /mnt/expand: LSPosed starts before that storage is mounted and never loads the Settings tile's hook ("Failed to find package info"). Move it to internal storage, then reboot.

Tap Move to internal storage, then reboot.

4.6 Speaker tab: the Status card

The Status card lists what the module reports. Values in the accent colour are fine; values in red need a look.

Row	Values	Meaning
Armed	yes / no	Whether the earpiece is switched on

Daemon	running / stopped	Whether the module's background program runs
Playback	active / idle	Whether the module sees audio playing
Mode	playback / always	When the routing is applied (see Mode below)
Channels	mono, stereo, or e.g. mono -> stereo after reboot	The running channels, and a pending switch (module v0.9.0+)
Nothing OS	new build: stereo unavailable or not a CMF Phone 1	Shown only when the Nothing OS build is not the known one
Routing	e.g. 3 actions	How many routing actions are configured; red at 0
Earpiece gain	e.g. 8 dB, unreadable, or 31 = mute (update the module)	The configured earpiece gain
Guard	active / bypassed	Whether the earpiece follows the speaker amp
Module	e.g. v1.0.9-beta	The installed module version
App	e.g. 1.0.97-beta	This app's version

> TECHNICAL
>
> The card is filled from one root shell: `stereoctl status brief` (modules before v0.9.3 ignore `brief` and print the full status), `stereoctl --help` (to see which commands the module supports), the `Handset Volume` line of `actions.conf`, `module.prop`, the `guard_bypass` file, `stereo.conf` and the property `vendor.audiohal.cmf_prot`.

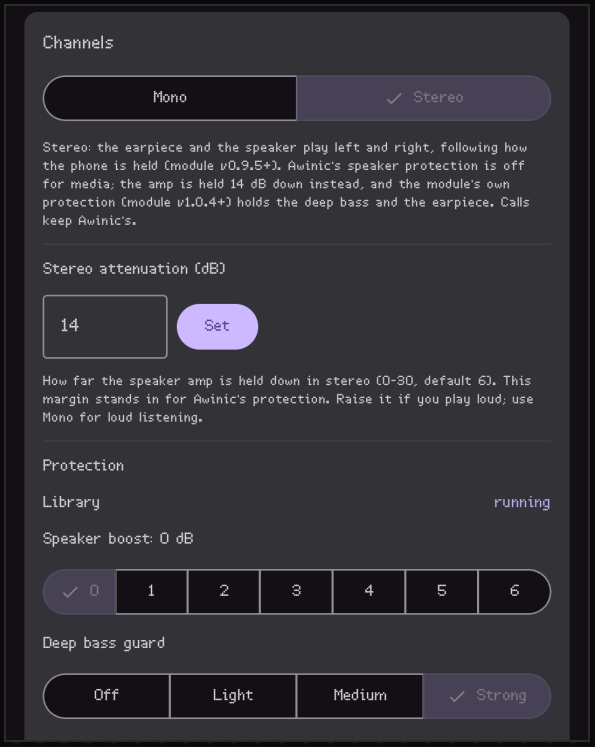
4.7 Speaker tab: Channels

The Channels card has a two-part button, Mono and Stereo, with the chosen one selected. The text under it explains the choice:

- Mono: Mono: stock audio. Both drivers play the full mix, with Awinic's speaker protection.
- Stereo: Stereo: the earpiece and the speaker play left and right, following how the phone is held (module v0.9.5+). Awinic's speaker protection is off for media; the amp is held 6 dB down instead, and the module's own protection (module v1.0.4+) holds the deep bass and the earpiece. Calls keep Awinic's. (The number is your current stereo attenuation.)

Tapping Mono switches at once. Tapping Stereo first shows the Switch to stereo? dialog.

The card only appears with module v0.9.0 or newer.



The Channels card in Stereo, here with the attenuation raised to 14 dB, and the Protection block.

4.8 The Switch to stereo? dialog

Pro explains in detail what Stereo changes before it switches:

Stereo switches off two mono folds: one in MediaTek's audio HAL, and Awinic's speaker-protection processing on the audio DSP. The phone's own vendor files are patched in the module overlay after hash checks; anything unrecognised is refused.

Without Awinic's processing, media on the speaker relies on the amp's hardware cut-outs, a fixed attenuation and the module's own protection (deep bass guard, earpiece limiter). Use Mono for loud listening and back off at the first distortion.

Applied at once by restarting the audio HAL (a moment of silence; players pause). If that cannot be done safely - no verified files, a call in progress, or the HAL does not come back - it is undone and waits for a reboot. Two failed boots put it back to mono automatically.

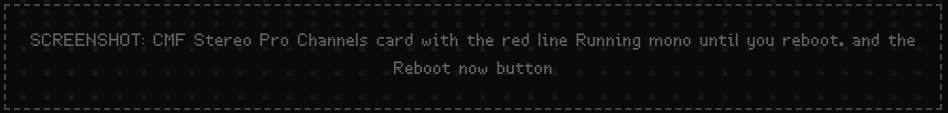
Tap **Switch to stereo** or **Cancel**.

SCREENSHOT: CMF Stereo Pro dialog Switch to stereo? with its three paragraphs and the buttons Switch to stereo and Cancel

4.9 Running mono until you reboot: Reboot now

If a switch could not be applied live, the Channels card says in red, for example, Running mono until you reboot., with a **Reboot now** button. The Status card's Channels row shows the same, e.g. mono -> stereo after reboot.

Tap **Reboot now**. The dialog **Reboot now?** says *The phone restarts to switch to stereo.* Tap **Reboot** or **Cancel**.



4.10 Stereo attenuation (dB)

When Stereo is chosen, the Channels card has a field **Stereo attenuation (dB)** and a **Set** button. This is how far the speaker amp is held down in Stereo: the bigger the number, the quieter and gentler on the speaker.

Type a number from 0 to 30 (digits only, at most two) and tap **Set**. The default is 6. The text under the field reads: *How far the speaker amp is held down in stereo (0–30, default 6). This margin stands in for Awinic's protection. Raise it if you play loud; use Mono for loud listening.*

This is the same setting as CMF Stereo's SPEAKER LEVEL: 6 dB is level 10, 8 dB is level 9, and so on to 24 dB at level 1 (chapter 5). Pro lets you use any whole number in between, and above 24.

> TECHNICAL
>
> Writes `STEREO_ATTEN_DB` in `stereo.conf` (clamped to 0–30), reads it back, and reports *stereo attenuation -> N dB (applied on the daemon's next cycle)*. The daemon applies it through the amp's own volume control, `aw_dev_0_volume`, in 1/8 dB steps. An empty field disables Set; a value that is not a number is taken as 6.

4.11 The warning under 6 dB (new in v1.0.97-beta)

6 dB is the margin that stands in for Awinic's protection in Stereo. CMF Stereo never goes below it. Pro can, but now warns you first, in two ways.

A red line under the field. As soon as you type a value under 6, a red line appears:

- With the module's protection running: Under 6 dB the speaker is held only by the module's protection (its peak limiter). Risky for long loud listening.
- Without it: DANGER: the module's protection is not running, so under 6 dB nothing but the amp's own cut-outs protects the speaker.

A dialog before it is set. Tapping Set with a value under 6 opens a dialog instead of setting it. With the protection running it is titled **Lower the speaker's margin?**; without it, **Speaker unprotected.** Both begin: In stereo, Awinic's speaker protection is off for media. The 6 dB attenuation is the margin that stands in for it; CMF Stereo never goes below it. Then:

- Protection running: The module's protection is running: its limiter still holds the speaker's peaks where Awinic would in mono. But Awinic's thermal protection is off in stereo, so long loud listening at N dB can overheat and damage the speaker.
- Not running: The module's protection is NOT running (after a live switch from a mono boot it only starts after a reboot). At N dB only the amp's hardware cut-outs protect the speaker: loud audio can damage it. Reboot in stereo first, or keep 6 dB or more.

Both end: Back off at the first distortion; use Mono for loud listening. The buttons are **Set N dB anyway** (in red) and **Cancel**.

> WARNING
>
> Values under 6 dB are for short tests only. Never use them without the protection running, and never for long, loud listening.

SCREENSHOT: CMF Stereo Pro Stereo attenuation field with 3 typed and the red warning line under it (protection running)

SCREENSHOT: CMF Stereo Pro dialog Lower the speaker's margin? with Set 3 dB anyway in red and Cancel

SCREENSHOT: CMF Stereo Pro dialog Speaker unprotected (protection not running, after a live switch from a mono boot) with Set N dB anyway and Cancel

> TIP
>
> This warning is new in v1.0.97-beta and has not been seen on a phone yet.

4.12 Protection block

When Stereo is chosen and the module has protection (v1.0.4+), the Channels card continues with **Protection**:

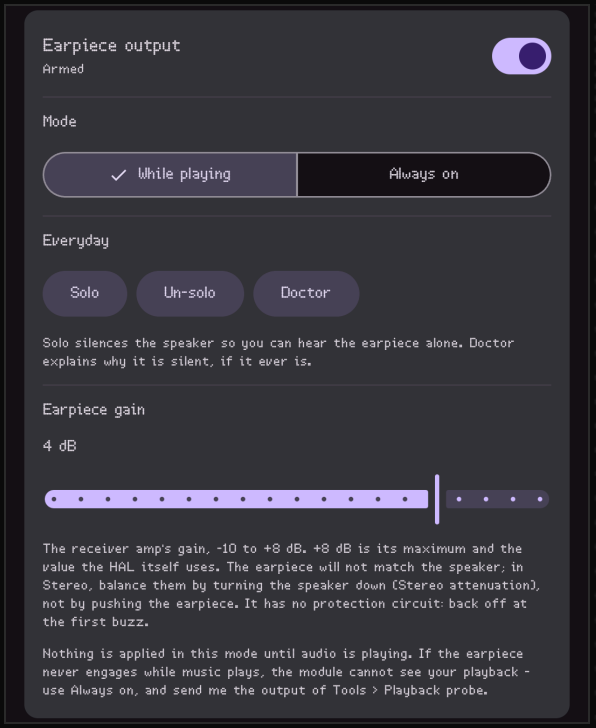
- **Library**: running, or not running yet in red. This is whether the module's protection library is active in the audio service.
- **Speaker boost**: N dB, with seven buttons 0 to 6.
- **Deep bass guard**: Off, Light, Medium, Strong.
- **Earpiece leveller** (module v1.0.5+): Off, Light, Medium, Strong, with the text Raises the earpiece's quiet and medium passages towards -18 dBFS of loudness by at most 4 / 8 / 12 dB (over 400 ms; down within 10 ms), leaves passages below -60 dBFS alone, changes no frequency, and keeps peaks at the earpiece's ceiling (-0.1 dBFS with the limiter off).
- **Earpiece limiter (-2.5 dBFS)**: a switch.

At the bottom, the measured basis: Measured with DRIVER STUDY: Awinic in mono holds the speaker's peaks about 5 dB below the amp's full output and cuts 100 Hz by 16 dB at any level. The boost lowers the fixed attenuation by up to 6 dB, only while the library runs, with a limiter that keeps the peaks at Awinic's level. The bass guard holds the band below 100 Hz at -8 / -14 / -20 dB below full output, only when it is louder. The earpiece clips in its top 1.5 dB; its limiter holds it at -2.5 dBFS.

These are the same settings as in CMF Stereo; chapter 5 explains them.

4.13 Speaker tab: Earpiece output

The next card starts with the **Earpiece output** switch, with Armed or Disabled under it. It is the same as CMF Stereo's EARPIECE SPEAKER: on plays media through the earpiece too; off stops the module's daemon and restores the controls it touched.



The Earpiece output card: switch, Mode, Solo / Un-solo / Doctor and the earpiece gain.

4.14 Speaker tab: Mode

Mode decides when the earpiece routing is applied:

- **While playing** (the default): only while audio plays. Kinder to the earpiece, which is not held energised all day.
- **Always on**: permanently. Harder on the earpiece, and it costs idle power.

If the mode is While playing and nothing is playing, the card adds: Nothing is applied in this mode until audio is playing. If the earpiece never engages while music plays, the module cannot see your playback - use Always on, and send me the output of Tools > Playback probe. (Playback probe is on the Developer tab.)

> TECHNICAL
>
> Sets `MODE=playback` or `MODE=always` in `stereo.conf`. CMF Stereo has no Mode setting; it uses whatever the module is set to.

4.15 Speaker tab: Everyday (Solo, Un-solo, Doctor)

- **Solo**: silences the speaker so you hear the earpiece alone. Useful to check that the earpiece really plays, or to judge it.
- **Un-solo**: brings the speaker back.
- **Doctor**: explains why the earpiece is silent, if it ever is. Disabled on modules that do not have it (it needs module v0.7.1+).

The card says: Solo silences the speaker so you can hear the earpiece alone. Doctor explains why it is silent, if it ever is.

> TECHNICAL
>
> Solo runs `stereoctl solo`, which bypasses the guard first (otherwise turning the amp off would read as "headphones took over" and silence the earpiece too). Doctor runs `stereoctl doctor`: it walks the six things that can cause silence (kill switch, dead daemon, unusable tinymix, idle playback, a blocking guard, a control that will not take its value) and names the one responsible. It deliberately does nothing when playback is idle, so play something first.

4.16 Speaker tab: Earpiece gain

The earpiece's gain in decibels, on a slider from -10 (left, quietest) to +8 dB (right, loudest) in 1 dB steps. The value is shown above the slider. Let go of the slider to set it.

The card says: The receiver amp's gain, -10 to +8 dB. +8 dB is its maximum and the value the HAL itself uses. The earpiece will not match the speaker; in Stereo, balance them by turning the speaker down (Stereo attenuation), not by pushing the earpiece. It has no protection circuit: back off at the first buzz.

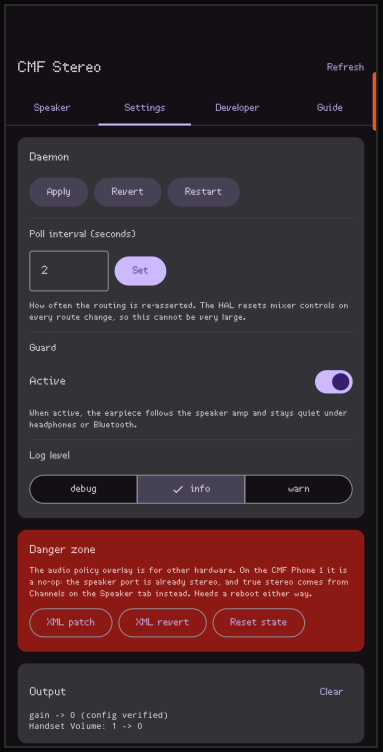
If the gain cannot be read, the slider is disabled and shows unavailable, with the red note: Could not read the gain from actions.conf, so the slider is disabled rather than risk writing a wrong value over it.

> TECHNICAL
>
> Writes `Handset Volume` in `actions.conf` (register value = 8 - dB, clamped to 0-18), reads it back, then applies it to the mixer at once with `stereoctl ctl 'Handset Volume' N` so you hear it immediately. The Output card shows `gain -> N (config verified)`. CMF Stereo's EARPIECE LEVEL sets the same value in 2 dB steps.

4.17 Settings tab: Daemon

The Daemon card controls the module's background program:

- **Apply**: applies the configured routing once.
- **Revert**: restores the values saved before the module touched them.
- **Restart**: restarts the daemon, for example after you edited the config by hand.



The Settings tab: the Daemon card and the Danger zone.

4.18 Settings tab: Poll interval

Poll interval (seconds): how often the routing is re-asserted. Type a whole number and tap **Set**. The module's default is 2.

The card explains: How often the routing is re-asserted. The HAL resets mixer controls on every route change, so this cannot be very large.

> TECHNICAL
>
> Writes `WATCH_INTERVAL` in `stereo.conf`.

4.19 Settings tab: Guard

The **Guard** switch shows **Active** or **Bypassed**. When active, the earpiece follows the speaker amp and stays quiet under headphones or Bluetooth. Bypass it only if you want the routing applied regardless.

> TECHNICAL
>
> Runs `stereoctl guard on` or `guard off`. On the CMF Phone 1 the guard is `aw_dev_0_switch = Enable`: the audio HAL disables the speaker amp when headphones or Bluetooth take over, so the earpiece stops with it.

4.20 Settings tab: Log level

Three buttons, **debug**, **info** and **warn**, set how much the daemon writes to its log. Use debug while chasing a problem, then go back to info.

> TECHNICAL
>
> Writes `LOG_LEVEL` in `stereo.conf`. Read the log on the Developer tab, Config files, Log.

4.21 Settings tab: Danger zone

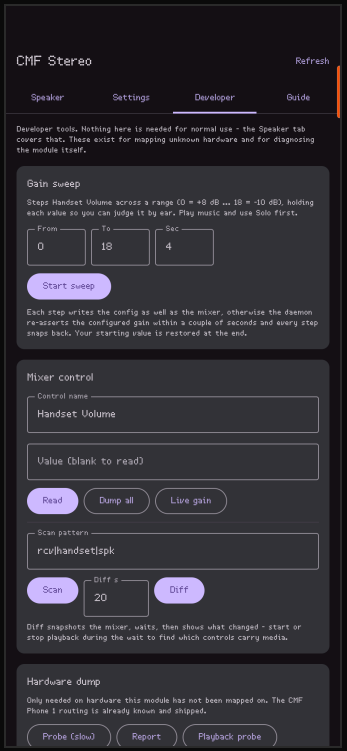
A red card with tools you should not normally need. It explains: The audio policy overlay is for other hardware. On the CMF Phone 1 it is a no-op: the speaker port is already stereo, and true stereo comes from Channels on the Speaker tab instead. Needs a reboot either way.

- **XML patch**: installs the audio policy overlay that widens the speaker port to stereo.
- **XML revert**: removes it.
- **Reset state** (modules with `reset`, v0.7.1+): clears the saved original values, the guard bypass, the solo state and the kill switch, then restarts. The saved originals survive reinstalls, so a bad one would make every revert restore a wrong value; this is the way out.

4.22 Developer tab: overview

The Developer tab opens with: Developer tools. Nothing here is needed for normal use - the Speaker tab covers that. These exist for mapping unknown hardware and for diagnosing the module itself.

Its cards: Gain sweep, Mixer control, Hardware dump, Run any stereoctl command, and Config files. All output appears in the Output card.



The Developer tab: Gain sweep and Mixer control.

4.23 Gain sweep

Steps the earpiece gain across a range, holding each value so you can judge it by ear. Play music and use Solo first.

1. Fill in From and To (register values, default 0 and 18) and Sec (seconds per step, default 4).
2. Tap Start sweep. The card shows Now playing gain: N in large type.
3. Tap Stop to end early.

At the end, your starting value is restored, and the Output card lists every step with Sweep done. Set whichever value sounded best on the Speaker tab.

> TIP
>
> The sweep counts in register values, where 0 is the loudest: 0 = +8 dB, 8 = 0 dB, 18 = -10 dB. To set the value you liked on the Speaker tab's slider, take dB = 8 minus the value.

> TECHNICAL
>
> Each step writes the config as well as the mixer; otherwise the daemon re-asserts the configured gain within a couple of seconds and every step snaps back. To is capped at 18.

4.24 Mixer control, Dump all and Live gain

Read or write one of the phone's audio mixer controls.

- **Control name** (default `Handset Volume`) and **Value** (blank to read).
- The button says **Read** when Value is empty and **Write** when it is filled.
- **Dump all**: every mixer control and its current value (about 1400 lines on this phone).
- **Live gain**: the earpiece gain the mixer holds right now, as opposed to what `actions.conf` says it should be.

> WARNING
>
> Writing mixer controls by hand can silence the audio, crash the audio service or drive the earpiece past what it is built for. Write only what you understand.

> TECHNICAL
>
> These run `stereoctl ctl NAME [VALUE]` and `stereoctl dump`.

4.25 Scan and Diff

- **Scan pattern** (default `rcv|handset|spk`) and **Scan**: searches the mixer for controls matching the pattern.
- **Diff s** (default 20) and **Diff**: Diff snapshots the mixer, waits, then shows what changed - start or stop playback during the wait to find which controls carry media. This is how the media path on this phone was found in the first place.

> TECHNICAL
>
> `stereoctl scan [pattern]` and `stereoctl diff [seconds]`.

4.26 Hardware dump

Only needed on hardware this module has not been mapped on. The CMF Phone 1 routing is already known and shipped.

- **Probe (slow)**: a full hardware dump (audio topology, all mixer controls, amplifier drivers, audio service configs, MediaTek's parameter tree). Takes a minute or two.
- **Report**: folds the newest probe into one shareable text file, `/sdcard/cmf-stereo-report.txt`.
- **Playback probe**: shows what the module can see of playback (the state of every playback stream, and what the module concludes). Use it when the earpiece never engages in While playing mode.

4.27 Run any stereoctl command

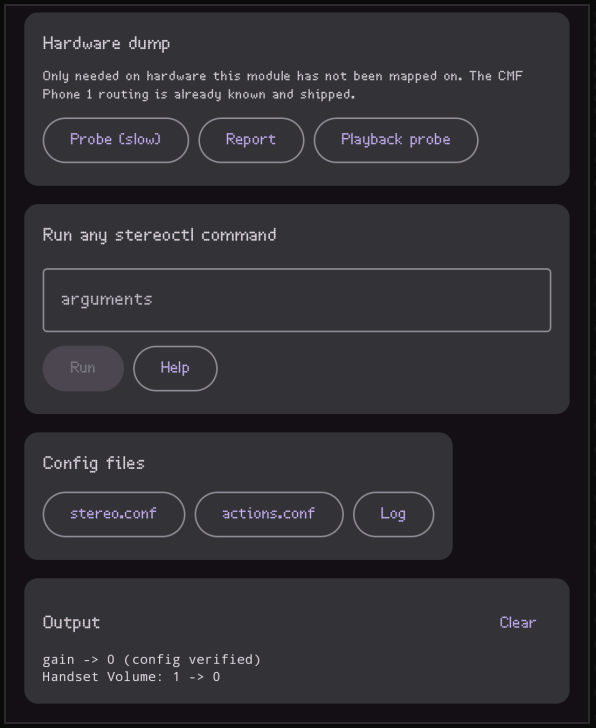
Type any `stereoctl` arguments in **arguments** and tap **Run**, for example `protect` or `channels`. **Help** shows the module's full command list.

> WARNING
>
> The command runs as root. Type only commands you understand.

4.28 Config files

Three buttons show the module's files in the Output card:

- **stereo.conf**: the module's behaviour settings (mode, channels, attenuation, protection, poll interval, log level).
- **actions.conf**: the routing itself, one action per line.
- **Log**: the last 60 lines of the daemon's log.

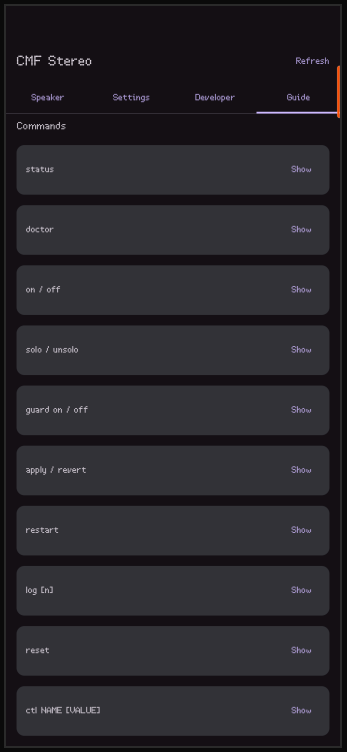


Further down: Hardware dump, any stereoctl command, the config files and the Output card.

4.29 Guide tab

A built-in reference, written from what was measured on the CMF Phone 1. Type in Search the guide to filter it; while you search, the matching entries open by themselves. Otherwise tap Show on an entry to open it and Hide to close it. If nothing matches, it says Nothing matches "…".

Section	Entries
Start here	What this module does; Mono or stereo; Everyday use; If the earpiece goes silent
Commands	status; doctor; on / off; solo / unsolo; guard on / off; apply / revert; restart; log [n]; reset; ctl NAME [VALUE]; scan [pattern]; dump; diff [seconds]; probe [dir]; report; channels [mono|stereo]; xml-patch / xml-revert
Settings (stereo.conf)	MODE; PLAYBACK_CTL / PLAYBACK_VALUE; GUARD_CTL / GUARD_VALUE; WATCH_INTERVAL; CHANNELS; STEREO_ATTEN_DB; LOG_LEVEL; actions.conf syntax
This device	How the earpiece is driven; Where the mono fold is; Earpiece gain (Handset Volume); Safety; Recovery



The Guide tab: every entry opens with Show.

05

LEVELS AND PROTECTION EXPLAINED

This chapter explains what the levels and the protection settings do underneath, with the exact numbers, and how to use them well and safely.

5.1 Earpiece level and Earpiece gain

CMF Stereo's EARPIECE LEVEL (1 to 10), Pro's Earpiece gain slider (-10 to +8 dB) and the Nothing Settings page's Earpiece level all set the same thing: the earpiece amplifier's gain, called Handset Volume.

Level	Gain	Handset Volume
10	+8 dB (maximum)	0
9	+6 dB	2
8	+4 dB	4
7	+2 dB	6
6	0 dB	8
5	-2 dB	10
4	-4 dB	12
3	-6 dB	14
2	-8 dB	16
1	-10 dB	18

Pro also reaches the odd steps in between (+7, +5 dB and so on). CMF Stereo then shows the nearest level, rounding a half step to the quieter level.

TECHNICAL

Handset Volume is the receiver amp's gain on the phone's MT6369 codec. Its range is 0–18, and the register runs opposite to the dB scale: 0 = +8 dB (loudest), 8 = 0 dB, 18 = -10 dB, and 31 = -40 dB (the codec's mute value). The audio HAL itself sets 0 whenever it drives the receiver. Module versions before v0.9.2 shipped 31, believing it was the top; the module's installer now corrects any value above 18 to 0. Pro's Status card shows 31 = mute (update the module) if it still finds such a value. The digital gain before the DAC (ADDA_DL_GAIN) is left to the HAL, because it is shared with the headphone path.

5.2 Speaker level and Stereo attenuation

CMF Stereo's SPEAKER LEVEL (1 to 10), Pro's Stereo attenuation (dB) and the Nothing Settings page's Speaker level all set how far the speaker amp is held down in Stereo, called STEREO_ATTEN_DB.

Level	Attenuation
10	6 dB (the safety margin)
9	8 dB
8	10 dB
7	12 dB
6	14 dB
5	16 dB
4	18 dB

3	20 dB
2	22 dB
1	24 dB

Pro accepts any whole number from 0 to 30. Values under 6 show as level 10 in CMF Stereo, values over 24 as level 1; odd values show as the nearest level.

> TIP
>
> If Pro set a value under 6, CMF Stereo shows level 10 already lit, so tapping the tenth dot does nothing. To bring it back to the 6 dB margin from CMF Stereo, tap dot 9 and then dot 10, or set 6 in Pro.

> TECHNICAL
>
> The daemon holds the amp at least this many dB down through the amp's own volume control, `aw_dev_0_volume` (1/8 dB steps), less the speaker boost while the protection runs. It re-reads the value every cycle; no restart is needed. In Mono the amp gets its previous value back.

5.3 Balancing the two sides

The earpiece is much smaller than the bottom speaker and will not match it. To get an even stereo picture:

1. Leave the earpiece at level 10 (+8 dB), unless it buzzes.

2. Switch on the EARPIECE LIMITER (it is on by default).
3. Play something with clear left and right, and lower the SPEAKER LEVEL until the earpiece side sounds as clear as the speaker side.
4. If the earpiece's quiet passages get lost, try the EARPIECE LEVELLER at LOW or MID.

> WARNING
> Balance by turning the speaker down, never by pushing the earpiece. Lower is also gentler on the speaker.

> TIP
> The CMF L/R Test app (chapter 7) and the module's test files LEFT_ONLY_3min.wav and LEFT10s_RIGHT10s.wav, which come with every module release, help you hear which side is which.

5.4 Why 6 dB is the floor in CMF Stereo

In Mono, Awinic's protection holds the speaker's peaks about 5 dB below the amp's full output. Stereo switches that processing off for media, so the module holds the speaker 6 dB down instead. That 6 dB margin is what stands in for Awinic's protection, which is why CMF Stereo's loudest speaker level is exactly 6 dB and never less.

CMF Stereo Pro can go lower for testing, after the warning described in chapter 4.

5.5 The module's protection: overview

In Stereo, the module runs its own protection inside the phone's audio service, built from measurements of what Awinic does in Mono. It has four parts:

Setting	Protects	Default
Speaker boost	(makes the speaker louder, with a peak limiter)	0 (off)
Deep bass guard	The speaker, from loud deep bass	Medium (MID)
Earpiece limiter	The earpiece, from clipping	On
Earpiece leveller	(brings up the earpiece's quiet parts)	Off

The protection applies in Stereo only. In Mono, Awinic's own protection is on, and the earpiece plays the stock way, as it always did.

> TECHNICAL
>
> Needs module v1.0.4 (the leveller v1.0.5). Set with `stereoctl protect boost 0-6`, `protect bass off|light|medium|strong`, `protect earpiece on|off` and `protect leveller off|light|medium|strong`; stored in `stereo.conf` as `SPEAKER_BOOST`, `BASS_PROTECT`, `EARPIECE_LIMIT` and `EARPIECE_LEVELLER`. The daemon applies a change within one cycle.

5.6 When the protection runs

The protection library starts with the first sound on the speaker after the phone has booted in Stereo. CMF Stereo's SPEAKER BOOST hint says Starts with the next sound

on the speaker. until it runs; Pro's Library row says not running yet.

After a live switch from a Mono boot, the protection does not start until the next reboot: the library is not in place until then. Stereo still plays, held at the fixed attenuation, but the boost is ignored. Reboot once in Stereo to get the full protection.

> TECHNICAL
>
> The app treats the protection as running when the property vendor.audiohal.cmf_prot is 1 and the running channels are stereo. The module's README lists the protection after a live switch from a Mono boot as not yet tested on a phone.

5.7 Speaker boost

Makes the speaker louder by lowering the fixed attenuation by up to 6 dB, one step per dB. A limiter then keeps the speaker's loudest moments where Awinic keeps them in Mono, about 5 dB below the amp's full output. So the speaker gets louder on average, but its peaks do not.

- 0 is the sound you had before.
- It works only while the protection library runs.

> TIP
>
> Boost 6 has been tried only briefly; a longer listen is still to be tested.

5.8 Deep bass guard

Holds back only the deepest bass (below 100 Hz), and only when it gets loud enough to strain the speaker. Quiet bass is untouched.

CMF Stereo	Pro / Settings page	Deep bass held at
OFF	Off	not held
LIGHT	Light	-8 dB below full output
MID	Medium	-14 dB below full output
STRONG	Strong	-20 dB below full output

> TECHNICAL
>
> Measured: Awinic in Mono cuts 100 Hz by 16 dB at any level. Before the guard (module v1.0.4), Stereo let the speaker get up to about 10 dB more deep bass than Awinic does.

5.9 Earpiece limiter

Keeps the earpiece below the point where it crackles. Measurements showed the earpiece clips in its top 1.5 dB of full scale; the limiter holds it at -2.5 dBFS. It is on by default.

> WARNING
>
> Turning the limiter off removes the earpiece's only protection in Stereo.

5.10 Earpiece leveller

Brings up the earpiece's quiet parts so it keeps up with the speaker. Loud parts, bass and every frequency stay as they are, and the earpiece's peaks are never raised.

CMF Stereo / Settings page	Pro	Quiet passages raised by at most
OFF	Off	0
LOW	Light	4 dB
MID	Medium	8 dB
HIGH	Strong	12 dB

> TECHNICAL
>
> It raises the earpiece's quiet and medium passages towards -18 dBFS of loudness, rising over 400 ms and coming down within 10 ms, leaves passages below -60 dBFS alone, changes no frequency, and keeps peaks at the earpiece's ceiling (the limiter's -2.5 dBFS, or -0.1 dBFS with the limiter off). On the test phone it made a clear difference on the earpiece side, with no distortion heard.

5.11 Calls, headphones and Bluetooth

- **Calls** always keep Awinic's speaker protection, in Mono and Stereo, and are not affected by a Mono/Stereo switch.
- **Headphones and Bluetooth**: with the guard active (the default), the earpiece follows the speaker amp, so it stays quiet when headphones or Bluetooth take over.

5.12 The module's safety nets

Besides the protection, the module has safety nets that act on their own:

- **Boot watchdog**: two failed boots with the Stereo overlay in place remove it and set Mono.
- **Crash-loop guard**: if the audio service restarts 3 times within a minute in Stereo, the module puts the stock files back (Mono) at once and tells the apps.
- **Unknown Nothing OS build**: Stereo stays off (NOTHING OS UPDATED).
- **Hash checks**: the module patches only audio files whose checksums match the analysed build; anything else is refused and the phone stays in Mono.

> TIP
> These are covered by the module's tests on fake files; none has been needed on the phone yet.

06

QUICK SETTINGS AND NOTHING SETTINGS

You do not have to open the app to switch Stereo. Each edition adds a Quick Settings tile, and, with LSPosed, a Stereo speakers page inside Nothing's own Settings that you can also find with Settings search.

6.1 The Quick Settings tile

Each edition has a Quick Settings tile: **Stereo** for CMF Stereo, **Stereo Pro** for CMF Stereo Pro. It needs no LSPosed.

To add it: pull down the Quick Settings panel, open the tile editor, find Stereo or Stereo Pro, and drag it into your tiles.

SCREENSHOT: Quick Settings tile editor showing the Stereo (CMF Stereo) and Stereo Pro tiles with the dot-matrix speaker icon

6.2 Using the tile

- **Tap**: switches between Mono and Stereo. While the audio service restarts, the tile is greyed and says Switching….
- **Long press**: opens the app.

Tile	Subtitle	Meaning
On (lit)	Earpiece + speaker	Stereo is chosen
Off	Mono	Mono is chosen
Greyed	Switching…	A switch is in progress
Greyed	Unavailable	No root for the app, or the module is not installed

> **WARNING**
>
> Unlike the apps, the tile switches to Stereo straight away, without the warning dialog. Read the safety notes (chapter 1) before you rely on it.

> **TIP**
>
> The tile shows the chosen mode. If a switch could not be applied live and waits for a restart, the tile already shows the new mode; open the app to see that a restart is needed. The tiles' names and a tap on them have not yet been checked on a phone.

SCREENSHOT: Quick Settings panel with the Stereo tile lit, subtitle Earpiece + speaker

6.3 Stereo inside Nothing Settings: what it is

With LSPosed, each edition can also put Stereo inside Nothing's Settings:

- a **Stereo speakers** row in **Settings** > **Sound**, right above Dirac, drawn like Nothing's own rows;
- a **Stereo speakers** page inside Settings, built from Nothing's own rows, with the levels as dots and the choices as pills;
- the page's settings findable with **Settings** search.

This part is optional. The apps and the tile work without it.

> TECHNICAL
>
> Settings already supports rows from other apps (its `com.android.settings.action.EXTRA_SETTINGS` tile injection), but only takes them from system apps. The app's LSPosed hook marks only its own entry as a system one, then builds the page. Nothing else in Settings changes, and any failure leaves Settings as it was.

6.4 Setting up LSPosed

1. Install LSPosed and make sure the app is on internal storage.
2. Open LSPosed, go to Modules and enable one edition (CMF Stereo or CMF Stereo Pro). Its description reads Adds Stereo speakers to Nothing Settings > Sound. Scope: Settings.
3. In its scope, tick both **Settings** (`com.android.settings`) and **Settings Intelligence** (`com.android.settings.intelligence`, needed for Settings search).
4. Force-stop Settings, and force-stop Settings Intelligence once (or simply reboot).
5. Open Settings > Sound.

> WARNING
>
> Enable only one edition. With both enabled, Settings shows two Stereo speakers rows.

SCREENSHOT: LSPosed module scope screen for CMF Stereo with Settings and Settings Intelligence ticked

6.5 The Stereo speakers row in Sound

In Settings > Sound, the **Stereo speakers** row sits right above Dirac, as a plain row like Nothing's own (no icon, no divider). Under its title, a live summary shows the state:

Summary	Meaning
Stereo • left and right follow the screen	Stereo is running
Mono • both speakers play everything	Mono is running
Earpiece speaker off	The earpiece is switched off
Switches to stereo after a restart	A switch waits for a restart (or mono)
CMF true stereo module not installed	The module is missing
Needs root access for CMF Stereo	The app has no root

Tap the row to open the Stereo speakers page.

SCREENSHOT: Nothing Settings > Sound with the Stereo speakers row above Dirac and its summary Stereo • left and right follow the screen

6.6 The Stereo speakers page

The page opens inside Settings, with the toolbar title STEREO SPEAKERS, and every row has a description under its title. It has four groups:

Group	Row	Control	Description
Speakers	Earpiece speaker	Switch	Play through the earpiece as well as the bottom speaker
Speakers	Stereo	Switch	Changes with the state (next section)
Levels	Earpiece level	10 dots	10 is the loudest the earpiece goes. Turn it down if it ever buzzes
Levels	Speaker level	10 dots	Balances the two sides: lower it until the earpiece is as clear as the speaker
Protection	Speaker boost	7 dots	Louder speaker without more strain, +1 dB a dot. The first dot is off
Protection	Deep bass guard	Pills: Off, Light, Medium, Strong	Holds back only the deepest bass, and only when it gets loud enough to strain the speaker
Protection	Earpiece leveller	Pills: Off, Low, Mid, High	Brings up the earpiece's quiet parts so it keeps up with the speaker
Protection	Earpiece limiter	Switch	Stops the earpiece crackling at its very loudest
More	CMF Stereo	Tap	Open the app

One tap on a dot or pill sets it. The values and their meaning are the same as in the app (chapter 5).

SCREENSHOT: Nothing Settings Stereo speakers page, top: toolbar STEREO SPEAKERS, Speakers group with Earpiece speaker and Stereo switches, Levels group with dots

SCREENSHOT: Nothing Settings Stereo speakers page, scrolled: Protection group with Speaker boost dots, Deep bass guard and Earpiece leveller pills, Earpiece limiter, and More > CMF Stereo Open the app

6.7 The Stereo row's summary and greyed rows

The Stereo switch's description tells you the state:

- Left and right follow the screen: Stereo is chosen.
- Both speakers play everything: Mono is chosen.
- Switches after a restart: a switch waits for a restart.
- Switching…: a switch (or an earpiece on/off) is in progress; both switches are held until it is done.
- CMF Stereo needs root access, The CMF true stereo module is not installed, or Not available on this Nothing OS version yet: the page cannot be used, and every row is greyed.

Rows are also greyed when they do not apply: Speaker level and the protection rows only work while Stereo is running, and Earpiece leveller only with a module that has it. A greyed row's title is dimmed too.

> TIP
> When the page opens, its rows stay greyed for a moment until the real state has loaded; taps are ignored until then. That is deliberate: it is what stops the page from changing anything just by being opened.

> WARNING
> Like the tile, the page's Stereo switch has no warning dialog.

6.8 Always in step

The row, the page, the tile and both apps all show the same state. A change made in any of them, or outside them (the module's Action button, a `stereoctl` command in a terminal, or the module switching itself back to Mono) appears on the page while it is open and in the Sound row's summary, without a root call each time.

> TECHNICAL
>
> The app keeps the last state it read (StateCache). Each reading is published to Settings through the app's provider; when something changed, it notifies `content://<app>.settingstile/state` (the page listens to it) and the Sound row's summary address. Settings itself has no root: it reads the stored state instantly, and the app reads afresh in the background when the stored state is older than five minutes. Changes made outside the apps reach it because the module (v1.0.6+) calls the provider's `stateChanged` method after each change. The page updating by itself after an outside change has not yet been seen on a phone.

6.9 Settings search

With Settings Intelligence in the LSPosed scope, type a setting's name in Nothing's Settings search, for example stereo, earpiece, speaker, earpiece level or bass guard. The Stereo speakers rows are listed, with Stereo speakers under each result. Tap one to open the Stereo speakers page inside Settings, at that row.

With both editions installed, the rows are listed once (CMF Stereo's).

SCREENSHOT: Nothing Settings search for bass guard, listing Deep bass guard with Stereo speakers under it

TECHNICAL

The rows come from the app's search provider (authority <package>.search, protected by READ_SEARCH_INDEXABLES, which only Settings Intelligence holds), with the keywords stereo, speakers, earpiece, CMF and the row's group. The hook in Settings Intelligence accepts this provider, adds a marker to its list of indexed providers so it reads them again, and turns a tap on a result into Settings' own search trampoline with the page and the row's key. Settings Intelligence re-reads providers only when that list changes, which is why v1.0.80 showed nothing; fixed in v1.0.81.

6.10 If Nothing Settings changes

The page is built from parts of Nothing's Settings app whose internal names are checked every time Settings starts. If a Settings update no longer matches, nothing is hooked: the Stereo speakers row still appears and opens the app instead of the page. Nothing in Settings breaks.

TECHNICAL

The page was built against Nothing Settings 16.0.0. In the LSPosed log, look for CMF Stereo: settings page ready and CMF Stereo: settings search ready (3 hooks), or for settings page not installed (Settings differs), settings hook not installed or settings search not installed with the reason.

6.11 Without LSPosed

Without LSPosed (or with the app not enabled in it), Settings does not show the Stereo speakers row at all, because it only takes such rows from system apps. Settings search does not list the rows either. The apps and the Quick Settings tile work exactly the same.

07

CMF L/R TEST

CMF L/R Test is a small companion app that checks left and right. It plays a beep on the left channel only, in each of the ways apps hand sound to the phone, so you can hear that the beep always comes from the side on your left. It also has tools that record what the audio driver receives, used to develop the module.

7.1 What it is for

On the CMF Phone 1, Nothing's sound tuning (Dirac) has different tunings for music, films, speech and games, and picks one from the app that plays. Apps also hand their sound to Android in different ways (plain, low-latency, floating point, compressed). CMF L/R Test plays a left-only beep through each, so you can check that Stereo keeps left and right correct in every case, in both sideways positions.

It is a test tool, not something you need every day.

7.2 Installing it

Each release includes CMF-LR-test-<commit>.apk. Install it like any app; it appears as CMF L/R Test, with a crash-test-target icon (yellow and black quarters on a dot grid).

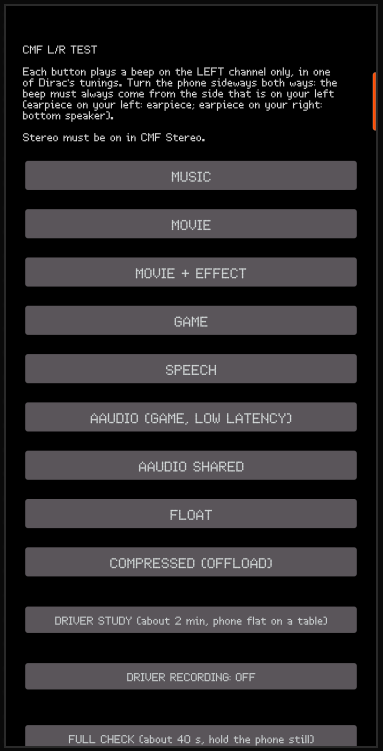
- Stereo must be on (in CMF Stereo, Pro, the tile or the module) for the left/right tests to mean anything.
- Root is used to read Dirac's tuning, the rotation the module stored, and which output Android plays through, and for the recording tools. Without root the test screens say (no root) and the recordings say Root refused.
- DRIVER STUDY asks for the microphone permission.

> TECHNICAL
>
> Package com.tiimoo.cmfstereo.modetest. Permissions: MODIFY_AUDIO_SETTINGS (to turn on the audio HAL's own recording) and RECORD_AUDIO (DRIVER STUDY).

7.3 The main screen

The main screen says: CMF L/R TEST. Each button plays a beep on the LEFT channel only, in one of Dirac's tunings. Turn the phone sideways both ways: the beep must always come from the side that is on your left (earpiece on your left: earpiece; earpiece on your right: bottom speaker). Stereo must be on in CMF Stereo.

Below it are the test buttons, then three tools: DRIVER STUDY, DRIVER RECORDING and FULL CHECK.



CMF L/R Test: each button plays a left-only beep in one of Dirac's tunings.

7.4 The tuning tests

Button	What it tests
MUSIC	Dirac's music tuning (Nothing's default)
MOVIE	Dirac's film tuning
MOVIE + EFFECT	The film tuning, with an audio effect attached to the app's own sound, as VLC does
GAME	Dirac's game tuning
SPEECH	Dirac's speech tuning (navigation)

> TECHNICAL
>
> Nothing's audio service picks the tuning from the name of the process that plays, matched against built-in lists (music, video, navigation, game, in that order). Each test runs in its own process, named to land in one tuning: :music, org.videolan.vlc.cmftest, org.videolan.vlc.cmftest2 (with a neutral DynamicsProcessing effect on its own session), com.google.android.apps.maps.cmftest, com.ludo.king.cmftest. These are only process names inside this app.

7.5 The playback-path tests

Button	What it tests
AAUDIO (GAME, LOW LATENCY)	Native AAudio, low latency and exclusive, as a game asks

AAUDIO SHARED	Native AAudio in shared mode
FLOAT	32-bit floating-point samples
COMPRESSED (OFFLOAD)	An MP3 of the same beep, decoded by the audio DSP, as music players do to save battery

Each screen shows what Android granted, for example whether the sound was offloaded.

> TIP
>
> On the CMF Phone 1, COMPRESSED (OFFLOAD) shows Compressed: this phone refuses MP3 offload and plays nothing: Android's offload check says no on this phone, so apps never use that path here. That is expected.

7.6 Reading a test screen

A test screen plays a short beep (880 Hz, 0.3 s on, 0.3 s off) on the left channel while it is open, and shows once a second:

- <NAME> TEST and LEFT channel only;
- Dirac reports: the tuning Dirac is using;
- Screen rotation: 90 (earpiece on your left), 270 (earpiece on your right), 180 or 0 (upright);
- Module reads rotation: the rotation the module stored;
- for some tests, how the beep was played;

- Android plays it through: the output Android chose;
- The beep must come from the side on your LEFT. Back to stop.

The test screens turn with the phone between the two landscape positions.

SCREENSHOT: CMF L/R Test MUSIC TEST screen in landscape, earpiece on the left: Dirac reports, Screen rotation 90 (earpiece on your left), Module reads rotation, Android plays it through

7.7 Running a left/right check

1. Switch to Stereo.
2. Open CMF L/R Test and tap a test, for example MUSIC.
3. Hold the phone sideways with the earpiece on your left: the beep must come from the earpiece.
4. Turn it the other way, earpiece on your right: the beep must now come from the bottom speaker.
5. Press Back to stop, and repeat with the other buttons.

On the test phone, MUSIC, MOVIE, MOVIE + EFFECT, GAME, SPEECH, AAUDIO, AAUDIO SHARED and FLOAT all came out on the correct side.

> TIP
>
> In Mono the beep comes from both speakers: that is correct for Mono.

7.8 DRIVER RECORDING

A switch button that turns on the audio driver's own recording of what it receives, to see in the data which channel a left-only sound is on.

1. Tap DRIVER RECORDING: OFF. It becomes DRIVER RECORDING: ON and says Recording. Play a test now, then turn this off.
2. Play a test.
3. Tap it again to stop. It says Saved: … and Send cmf-driver-dump.tgz and cmf-driver-dump.txt.

The files are saved as /sdcard/cmf-driver-dump.tgz and /sdcard/cmf-driver-dump.txt.

> TECHNICAL
>
> It sets vendor.streamout.pcm.dump and vendor.streamout.dsp.dump with resetprop and sends the same key through AudioManager.setParameters, which is what makes the HAL start recording. Recordings are collected from /data/vendor/audiohal/audio_dump.

7.9 FULL CHECK

FULL CHECK (about 40 s, hold the phone still) runs MUSIC, MOVIE, GAME and SPEECH one after another for 6 seconds each, with the driver recording on, and logs Dirac's tuning and the rotation each second.

1. Hold the phone sideways and tap FULL CHECK. It says Running: MUSIC, MOVIE, GAME, SPEECH… keep the phone still.

2. When it is done it says Saved: … and Turn the phone over and run it again, then send both files.

The file is named after the rotation: /sdcard/cmf-full-check-rot<rotation>.tgz.

7.10 DRIVER STUDY

A measurement used to design the module's protection. It plays tones on each driver, records them with the phone's microphones and the driver's own recordings, and writes the results to files.

Tap DRIVER STUDY (about 2 min, phone flat on a table). The study screen says: Put the phone flat on a table, screen up, in a quiet room, and do not touch it until it says done (about 3 minutes). It switches Stereo and Mono by itself (the sound stops for a moment at each switch) and puts back the mode you are in now. Media volume goes to maximum while it runs and back after: tones never louder than -3 dB below full scale. Tap START; allow the microphone when asked.

While it runs it shows the part, the tone and the progress, for example STEREO: playing, do not touch the phone. At the end: Done. Mode is … again. and the files to send.

- If the module does not answer, it stops: The CMF stereo module did not answer (stereoctl channels). Nothing was changed.
- Without the microphone permission: The microphone permission is needed for the study.

TECHNICAL

The screen is locked upright (no left/right swap). Stereo part: 300 ms tones on the earpiece, then the speaker, 11 frequencies from 100 Hz to 8 kHz at -30 to -3 dBFS. Mono part (stock files, Awinic on): tones on both channels, 7 frequencies, with the DSP's recordings of Awinic's input, output and current/voltage sense. Results: /sdcard/cmf-study.csv, /sdcard/cmf-study-mono.csv, /sdcard/cmf-study-notes.txt; raw recordings in /sdcard/cmf-study-raw.tgz. It checks after each switch that the mode really runs, and always switches back to the mode it found. Its rerun is no longer needed; the test stays in the app, its fixed alignment not yet run on a phone.

SCREENSHOT: CMF L/R Test DRIVER STUDY screen with its instructions and the START button

08

TROUBLESHOOTING

Find your problem by topic. Each entry says what you see, why, and what to do. If the answer is not here, see Reporting a problem at the end of this chapter.

8.1 The app shows a message instead of its controls

NEEDS ROOT / No root access. The app could not get root. Allow it in your root manager's Superuser list, then reopen CMF Stereo, or tap Refresh in Pro.

MODULE NOT INSTALLED / Module not installed. The module is missing or not active. Flash it, reboot, and check in your root manager that it is enabled.

READING… stays on screen. The root request may be waiting for an answer: look for your root manager's dialog.

8.2 The earpiece is silent

1. Check that **EARPIECE SPEAKER** (Earpiece output in Pro) is on.
2. If CMF Stereo says *Not running right now.*, tap **RESTART**.
3. Check that no headphones or Bluetooth device are connected: with the guard active, the earpiece stays quiet then.
4. In Pro, play some music and tap **Doctor** on the Speaker tab. It names the cause.
5. If the earpiece never engages while music plays in *While playing* mode, set Mode to **Always on** and run **Playback probe** on the Developer tab; send its output to the developer.
6. Use **Solo** to hear the earpiece alone and be sure.

8.3 Switching Mono and Stereo

It asks me to restart the phone. The module could not switch safely while the phone runs: no verified files, a call in progress, or the audio service did not come back. Your choice is saved; restart to apply it.

The music stopped when I switched. That is normal: the audio service restarts for about a second and players pause. Press play again.

There is no SOUND panel / Channels card. Your module is older than v0.9.0. Update it.

Stereo switched itself back to Mono. A safety net acted: two failed boots, the audio service restarting 3 times within a minute, or a Nothing OS update the module does not know. See NOTHING OS UPDATED below.

NOTHING OS UPDATED / Nothing OS: new build: stereo unavailable. The module does not know your new Nothing OS build yet and keeps Stereo off. Use Mono and update the module when a release supports the build.

Pro shows Nothing OS: not a CMF Phone 1. The module does not find a CMF Phone 1 audio stack; it does nothing on other phones.

8.4 Panels or settings are missing

Missing	Why	What to do

SPEAKER LEVEL and the protection panels (CMF Stereo)	Stereo is not running (Mono, or a switch waiting for a restart)	Switch to Stereo; restart if asked
Protection panels only	Module older than v1.0.4	Update the module
EARPIECE LEVELLER	Module older than v1.0.5	Update the module
EARPIECE LEVEL	The earpiece gain could not be read from the module's settings, or is out of range	Open Pro: its Status card shows unreadable or 31 = mute (update the module). Update the module, whose installer corrects an out-of-range value
Pro's Earpiece gain slider is disabled (unavailable)	Same cause; the app will not write a value it could not read	As above
Pro's Doctor is greyed	Module older than v0.7.1	Update the module

8.5 Protection not running

CMF Stereo's SPEAKER BOOST hint starts with Starts with the next sound on the speaker., or Pro's Library row says not running yet.

- If the phone booted in Stereo: play something on the speaker; the protection starts with the next sound.
- If you switched to Stereo live after a Mono boot: the protection only starts after a reboot. Reboot once in Stereo. Until then, the boost does nothing and the

speaker relies on the fixed attenuation and the amp's cut-outs.

> WARNING
> Without the protection running, never set Pro's stereo attenuation under 6 dB.

8.6 Sound quality and loudness

The earpiece buzzes, rattles or crackles. Lower EARPIECE LEVEL at once, and make sure EARPIECE LIMITER is on.

The speaker distorts in Stereo. Lower SPEAKER LEVEL, check that DEEP BASS GUARD is not OFF, reduce SPEAKER BOOST, or use Mono for loud listening.

The earpiece side is too quiet. Lower SPEAKER LEVEL to balance; try EARPIECE LEVELLER. Do not push the earpiece past level 10 by other means.

Left and right are reversed. Left and right follow the screen: with the earpiece on your left, the earpiece plays left. You need module v0.9.5 or newer. One known limit: an app with its own audio effect (VLC) and an ordinary app playing at the same time, earpiece on your right, can leave one of them reversed. Check with CMF L/R Test.

The first second of a stream sounds different. In Stereo, the amp's driver resets its slot on every stream start, so the first second or two plays as stock until the module's next cycle.

8.7 Full volume broken by another module

Other modules that change the phone's audio or volume can interfere with CMF True Stereo. A real case from the test phone: NothingTweaks' **Customise Volume Steps**, set to 24 steps, made the highest volume step silent, and the middle of the bar was the loudest, even with CMF True Stereo off. With the custom steps turned off, full volume was correct again.

If the sound behaves strangely and you run other audio or volume tweaks:

1. Turn off the other module's audio or volume feature, or disable that module, and reboot.
2. Check again with only CMF True Stereo active.
3. Switch other modules back on one at a time to find the one that clashes.

> WARNING
>
> Do not run the Hi-Res Audio module at the same time as CMF True Stereo: they touch the same files.

8.8 The Stereo speakers row is missing from Settings

1. Check that LSPosed is installed and working, and that the edition is enabled in it.
2. Check the scope: **Settings** (`com.android.settings`) must be ticked.
3. Check that the app is on internal storage. On adopted storage LSPosed cannot load it (its log says `Failed to find package info`); use MOVE TO INTERNAL STORAGE and

reboot.

4. Force-stop Settings (or reboot) after any change in LSPosed.
5. Look in the LSPosed log for lines starting with CMF Stereo:.

Two Stereo speakers rows. Both editions are enabled in LSPosed. Disable one, then force-stop Settings.

The row opens the app instead of the page. Your Settings app no longer matches what the page was built for (a Settings update); the LSPosed log says settings page not installed (Settings differs). Everything else still works.

The page's rows stay greyed. The page could not get the state: the app has no root, the module is missing, or Nothing OS is a build the module does not know. The Stereo row's description says which.

8.9 Settings search shows nothing

1. In LSPosed, make sure **Settings Intelligence** (com.android.settings.intelligence) is ticked in the app's scope, as well as Settings.
2. Force-stop **Settings Intelligence** once, or reboot.
3. Search for bass guard.
4. If both editions are installed, update both to v1.0.81 or newer. Before v1.0.81, an older edition could hide the newer one's rows, and rows indexed before the hook was on were skipped until the next version.

8.10 The Quick Settings tile

The tile says Unavailable. The app has no root, or the module is not installed. Open the app to see which.

The tile stays on Switching…. The audio service is restarting; wait a moment. If it stays, open the app to see the state.

Two tiles look alike. CMF Stereo's tile is called Stereo, Pro's Stereo Pro. If both say Stereo, update Pro (older versions used the same name).

8.11 Installing and updating problems

"App not installed" when updating. The installed app was signed with a different key: a build older than v1.0.35, or a debug build (your own or a fork's). Uninstall it once, then install the release.

The app will not install on the SD card. That is deliberate: both editions are internal-storage only.

The text is not in NDot. CMF Stereo found no NDot font on the phone and uses monospace. On a Nothing phone this should not happen.

The icon still shows a kinked outer wave. That edition is older than v1.0.76; update it (both editions get the same icon).

Many root toasts. Each root call can make your root manager show a toast. The app makes one root call per reading, and Settings uses the stored state without root.

8.12 The phone does not boot

Wait: after two failed boots with Stereo on, the module removes Stereo and the phone boots in Mono. If it still does not boot, disable the module by hand (chapter 2, If the phone does not boot), or boot into safe mode by holding volume down.

8.13 Reporting a problem

Report bugs as issues on the app's GitHub repository. Include, where you can:

- the app and module versions (Pro's Status card shows both), your phone model and Nothing OS build;
- what happens and the steps that cause it;
- Pro's Output card after Doctor, Playback probe or Log, or the module's log;
- for Settings problems, the LSPosed log lines starting with CMF Stereo:.

> TIP
> Security problems are reported privately, not as public issues: on the repository open the Security tab and choose Report a vulnerability. Only the latest release gets fixes, so update first and check that the problem still happens.

09

FREQUENTLY ASKED QUESTIONS

Short answers to the questions people ask most.

9.1 General

Do I need the app? No. The module works on its own; the app is a friendly way to control it. You can also switch Mono and Stereo with the Action button on the module's card in your root manager, or with `stereoctl`.

Which edition should I install? CMF Stereo for everyday use, Pro for tuning and diagnosing. You can install both.

Does the app run in the background? No. The module's daemon does the work; the app only reads and changes its settings when you use it (or when Settings, the tile or the module asks it to).

Does it work on other phones? The module is for the CMF Phone 1 (A015). On other phones it installs and does nothing.

Why is it called beta? CMF Stereo is still being tested; from v1.0.95-beta on, the versions say so.

Will I lose my settings when I update or uninstall the app? No. The settings belong to the module and live in `/data/adb/cmf-stereo/`.

9.2 Sound

Mono or Stereo: which is louder? Mono, with Awinic's full protection. Use Mono for loud listening.

Why does the speaker sound quieter in Stereo? It is held at least 6 dB down to keep it safe without Awinic's protection. SPEAKER BOOST makes it louder while keeping its peaks where Awinic keeps them.

Why doesn't the earpiece sound as full as the speaker? It is a small receiver built for calls. Balance the two by lowering the speaker.

Can I make the earpiece louder than level 10? No. Level 10 (+8 dB) is the receiver amp's maximum, the value the phone itself uses.

Do I have to rotate the stereo myself? No. Left and right follow the screen automatically.

Does Stereo affect calls? No. Calls keep Awinic's protection, and a Mono/Stereo switch is refused during a call.

What about headphones and Bluetooth? With the guard active (the default), the earpiece stays quiet when they take over.

9.3 Safety

Can Stereo damage my speaker? In Stereo the speaker has fixed limits but no temperature model. Use Mono for long, loud listening, lower the speaker level in Stereo, and back off at the first distortion.

Can it damage my earpiece? The earpiece has no protection circuit of its own. Keep the limiter on, and turn it down at the first buzz. Damage is permanent and would also affect calls.

Why can Pro go below 6 dB when CMF Stereo cannot? For tests. Pro warns you first and explains the risk.

What if Stereo goes wrong at boot? After two failed boots the module goes back to Mono by itself.

9.4 Nothing Settings and LSPosed

Do I need LSPosed? Only for the Stereo speakers row and page in Nothing Settings and for Settings search. The apps and the tile work without it.

Does the LSPosed part change anything else in Settings? No. It marks only its own row as a system one and builds its own page. If Settings changes, it hooks nothing.

Why does the page need Settings Intelligence in the scope? Settings Intelligence is the app behind Settings search. Without it the page still works, but search does not list its rows.

Can I enable both editions in LSPosed? You should not: Settings would show two rows.

10

TECHNICAL REFERENCE

Exact names, paths, commands and requirements, for readers who want to see what happens underneath or check something in a terminal.

10.1 Packages and components

App	Package	Label	Tile label
CMF Stereo	com.tiimoo.cmfstereo.clean	CMF Stereo	Stereo
CMF Stereo Pro	com.tiimoo.cmfstereo	CMF Stereo Pro	Stereo Pro
CMF L/R Test	com.tiimoo.cmfstereo.modetest	CMF L/R Test	(none)

Each edition contains:

- the app screen (also opened by a long press on the tile);
- a Settings entry, Stereo speakers, in category com.android.settings.category.ia.sound, order -61, key hint cmf_stereo;
- a provider <package>.settingstile (the row's summary, the page's state and changes);
- a search provider <package>.search, protected by android.permission.READ_SEARCH_INDEXABLES;
- the Quick Settings tile service;
- LSPosed module metadata: minimum Xposed API 93, scope com.android.settings and com.android.settings.intelligence.

Minimum Android version 8.0 (API 26); target API 35. Version names are 1.0.<build>-beta, the build number being the commit count.

10.2 Module files and paths

Path	What
/data/adb/modules/cmf_stereo	The installed module
/data/adb/modules/cmf_stereo/scripts/stereoctl	The command line the apps call
/data/adb/modules/cmf_stereo/module.prop	The module version
/data/adb/cmf-stereo/stereo.conf	Behaviour: mode, channels, attenuation, protection, poll interval, log level
/data/adb/cmf-stereo/actions.conf	The routing, one action per line (including ctl|Handset Volume|N)
/data/adb/cmf-stereo/stereo.log	What the daemon did
/data/adb/cmf-stereo/state/	Saved original values, used to revert
/data/adb/cmf-stereo/guard_bypass	Present while the guard is bypassed
/sdcard/cmf-stereo-report.txt	Written by Report

The configuration lives outside the module folder and survives module updates and reinstalls.

10.3 What each control runs

Control	Command or file change
EARPIECE SPEAKER / Earpiece output / Earpiece speaker	`stereoctl on / off`
MONO / STEREO, tile, Stereo switch	`stereoctl channels mono / stereo`
EARPIECE LEVEL, Earpiece gain	`Handset Volume` in `actions.conf`, read back, then `stereoctl ctl 'Handset Volume' N`
SPEAKER LEVEL, Stereo attenuation	`STEREO_ATTEN_DB` in `stereo.conf` (0–30), read back
SPEAKER BOOST	`stereoctl protect boost 0-6`
DEEP BASS GUARD	`stereoctl protect bass off|light|medium|strong`
EARPIECE LEVELLER	`stereoctl protect leveller off|light|medium|strong`
EARPIECE LIMITER	`stereoctl protect earpiece on|off`
RESTART (daemon), Restart	`stereoctl restart`
RESTART PHONE, Reboot now	`svc power reboot`, or `reboot`
MOVE, Move to internal storage	`pm move-package <package> internal`
Mode	`MODE=playback|always` in `stereo.conf`
Solo / Un-solo	`stereoctl solo / unsolo`
Doctor	`stereoctl doctor`
Apply / Revert	`stereoctl apply / revert`

Poll interval	WATCH_INTERVAL in stereo.conf
Guard	stereoctl guard on|off
Log level	LOG_LEVEL in stereo.conf
XML patch / XML revert / Reset state	stereoctl xml-patch / xml-revert / reset
Read / Write, Dump all, Live gain	stereoctl ctl NAME [VALUE], stereoctl dump, stereoctl ctl 'Handset Volume'
Scan, Diff	stereoctl scan [pattern], stereoctl diff [seconds]
Probe, Report	stereoctl probe, stereoctl report
Playback probe	playback states from /proc/asound/card*/pcm*p/sub*/status, and stereoctl status's playback line
Run, Help	stereoctl <arguments>, stereoctl --help
stereo.conf, actions.conf, Log	shows the files; stereoctl log 60

> TECHNICAL
>
> Commands are written to the root shell's standard input rather than passed with su -c, so the root shell parses exactly the text the app wrote (quoting survives). Each root call times out after 180 seconds.

10.4 stereo.conf settings

Key	Values	Default	Set from
MODE	playback, always	playback	Pro: Mode
CHANNELS	mono, stereo	mono	Mono/Stereo (set it through the switch, not by hand)
STEREO_ATTEN_DB	0–30	6	SPEAKER LEVEL, Stereo attenuation
SPEAKER_BOOST	0–6	0	SPEAKER BOOST
BASS_PROTECT	off, light, medium, strong	medium	DEEP BASS GUARD
EARPIECE_LIMIT	on, off	on	EARPIECE LIMITER
EARPIECE_LEVELLER	off, light, medium, strong	off	EARPIECE LEVELLER
WATCH_INTERVAL	seconds	2	Pro: Poll interval
LOG_LEVEL	debug, info, warn	info	Pro: Log level
PLAYBACK_CTL / PLAYBACK_VALUE	mixer controls separated by |	the DSP playback task switches	by hand
GUARD_CTL / GUARD_VALUE	a mixer control and value	aw_dev_0_switch = Enable	by hand

actions.conf syntax, one action per line, applied in order: ctl|<mixer control>|<value>, sysfs|<path>|<value>, prop|<property>|<value>. The original value of everything touched is

saved and restored on stop.

10.5 How the app reads the state

Everything the screen shows is read in one root shell (it used to take about fourteen). From `stereoctl status brief` the app takes:

- `armed: yes`, `daemon: running`, `playback: active`, `mode: …`, `(N configured)`;
- `channels: stereo`, or `channels: mono (switches to stereo after reboot)` for a pending switch;
- `build: known|updated|absent`.

From `stereoctl --help` it learns which features the module has: `doctor`, `reset`, `channels`, `protect` and `protect leveller`. Panels for features the module lacks are hidden.

A value that cannot be read is shown as unreadable, never as 0: a displayed 0 written back is how an early version once zeroed a working configuration.

10.6 Module version requirements

Feature	Module version
Doctor, Reset state	v0.7.1
Mono/Stereo (SOUND, Channels)	v0.9.0
Correct earpiece maximum (Handset Volume 0)	v0.9.2

Fast status reading (`status brief`), near-instant on/off	v0.9.3
Instant Mono/Stereo switch	v0.9.4
Left and right follow the screen	v0.9.5
NOTHING OS UPDATED notice; Action button	v0.9.8
Both speakers in games with While playing	v1.0.3
Protection: boost, deep bass guard, earpiece limiter	v1.0.4
Earpiece leveller	v1.0.5
Nothing Settings and tile follow outside changes	v1.0.6

This app release is made for module v1.0.6 or newer.

10.7 Nothing Settings internals

- **Row**: Settings' own tile injection. Settings takes such rows only from system apps (it checks `resolveInfo.system`); the hook on `queryIntentActivitiesAsUser` marks only this app's entry as system.
- **Page**: a tap on the row is sent to Settings' own `SubSettings` with a small Sound page (`MediaControlsSettings`) as carrier, titled Stereo speakers; once that page has built itself its rows are replaced with Nothing's own widgets (`NtCustSwitchPreferenceCompat`, `Preference`, `PreferenceCategory`), and the dots and pills are added under each row's summary in Settings' own colours and font.

- **State**: through the provider methods `getDynamicSummary`, `getPageState` and `setPage`, which answer only Settings and the app itself, on a thread of their own. `stateChanged` (called by the module) only triggers a fresh reading.
- **Search**: three hooks in Settings Intelligence: `PreIndexDataCollector.isWellKnownProvider`, `IndexDatabaseHelper.buildProviderVersionedNames` (marker `cmf-search:1`) and `IndexData$Builder.buildIntent` (results open Settings' `SEARCH_RESULT_TRAMPOLINE` with the row's key as `:settings:fragment_args_key`).

The obfuscated names in Settings were read from Nothing Settings 16.0.0 and are checked against their signatures before anything is hooked.

10.8 LSPosed log messages

Message	Meaning
CMF Stereo: settings page ready	The page is installed
CMF Stereo: settings page not installed (Settings differs): …	Settings does not match; the row opens the app
CMF Stereo: settings hook not installed: …	The row could not be let in
CMF Stereo: settings tile: …	An error while marking the row
CMF Stereo: settings page: …	An error inside the page (it carries on)
CMF Stereo: settings search ready (3 hooks)	Settings search is installed
CMF Stereo: settings search not installed: …	Settings Intelligence does not match

10.9 Privacy and permissions

- The apps ask for root through your root manager, and use it only to run the module's command and read or write its files.
- The apps' own manifest requests no Android permissions; there is no internet permission, so nothing is sent anywhere.
- The state is kept on the phone, in the app's private storage, so Settings can show it without root.
- The Settings provider answers only Nothing's Settings and the app itself; the search provider can only be read by Settings Intelligence.
- CMF L/R Test asks for MODIFY_AUDIO_SETTINGS and, for DRIVER STUDY, the microphone (RECORD_AUDIO). Its recordings are saved only to your phone's storage, for you to send if you choose.

> TIP
> Because the app holds root, the project treats as security problems: any way to make it run a root command you did not intend, exposing its root access to other apps, or writing values that push the speaker or earpiece past the documented ranges.

10.10 Signing and builds

Releases are signed with the project's permanent key (from v1.0.35). A local build without the key, or a build from a fork, is debug-signed and cannot update an installed

release. Release builds are shrunk, which is why each APK is about 1 MB (an early version was about 100 MB).

To build it yourself: `./gradlew assembleRelease`.

10.11 Version history

Version	Date	Change
v1.0.97-beta	8 Nov 2026	Pro warns before setting the speaker's stereo margin under 6 dB
v1.0.95-beta	8 Nov 2026	Versions marked beta; new banner
v1.0.81	8 Nov 2026	Settings search finds the stereo settings
v1.0.80	8 Nov	Stereo speakers settings in Settings search
v1.0.79	8 Nov	Bolder themed icon; CMF L/R Test gets its own icon
v1.0.76	7 Nov	Rounder outer sound wave on the icon
v1.0.75	7 Nov	Every text says left and right follow how you hold the phone
v1.0.71	6 Nov	Stereo in Nothing Settings (LSPosed), Quick Settings tile, instant sync without root, internal storage only
v1.0.60	5 Nov	Earpiece leveller (module v1.0.5)
v1.0.56	5 Nov	Speaker boost, deep bass guard, earpiece limiter (module v1.0.4)

v1.0.53	5 Nov	Both speakers in games (module v1.0.3); Nothing OS update notice
v1.0.39	4 Nov	Dot-matrix icon, live with Nothing2Something; licence CC BY-NC-SA 4.0
v1.0.35	4 Nov	Permanent signing key; instant Mono/Stereo; left and right follow the screen
v1.0.29	4 Nov	About 1 MB instead of about 100 MB; faster
v1.0.25	4 Nov	Two editions; Mono/Stereo; earpiece at its real maximum; speaker level; NDot font
v1.0.21	3 Nov	The first build of the two editions

10.12 Not yet seen on a phone

As of v1.0.97-beta, these features are built and checked in code but have not yet been seen on a real phone:

- Pro's warning before stereo attenuation under 6 dB (v1.0.97-beta);
- the Quick Settings tiles' names (Stereo / Stereo Pro) and a tap on them;
- the Stereo speakers page updating by itself while open after a change made outside it;
- the MOVE TO INTERNAL STORAGE panel;
- the NOTHING OS UPDATED panel.

Verified on the phone: both editions' on/off, live Mono/Stereo switching and levels; the protection and leveller settings; the Stereo speakers row and page in Nothing

Settings; the tiles in the tile editor; LSPosed listing the apps on internal storage; CMF L/R Test; Settings search.



CREDITS AND LICENCE

Who made CMF Stereo, and the terms you may use it under.

11.1 Credits

- **Dr.Essence**: author and maintainer. Ideas, design decisions, and every test on a real phone (logs, recordings, measurements) that the work rests on.
- Code written with Claude Code.
- Shout-out to **Komi Store**, the free, open-source app store for GitHub releases that keeps the app updated.
- The app is the front end for the **CMF True Stereo** module, in the same spirit as PerfMTK and its separate PerfMTK-Manager.

The module thanks **Adinata (@Adivenxnataly)**, whose Hi-Res Audio module pointed the project at the right part of MediaTek's audio stack, and the people who worked out dual-speaker mods on other phones and wrote them up.

11.2 Licence

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Releases up to v1.0.35 were published under OBSD; copies obtained under it keep those terms.

CMF Stereo draws Nothing's NDot font from the phone at runtime; no Nothing font ships in the app. The module bundles a tinymix binary built from tinyalsa (BSD-3-Clause).

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